



"AUSER" VENUS FLOATING CITY

1st January 2065

Orbital industry



1 Executive Summary

Position:

50km above the surface
of Venus.

8 Gas-turbine engines are used to
maintain its position.

Capacity:

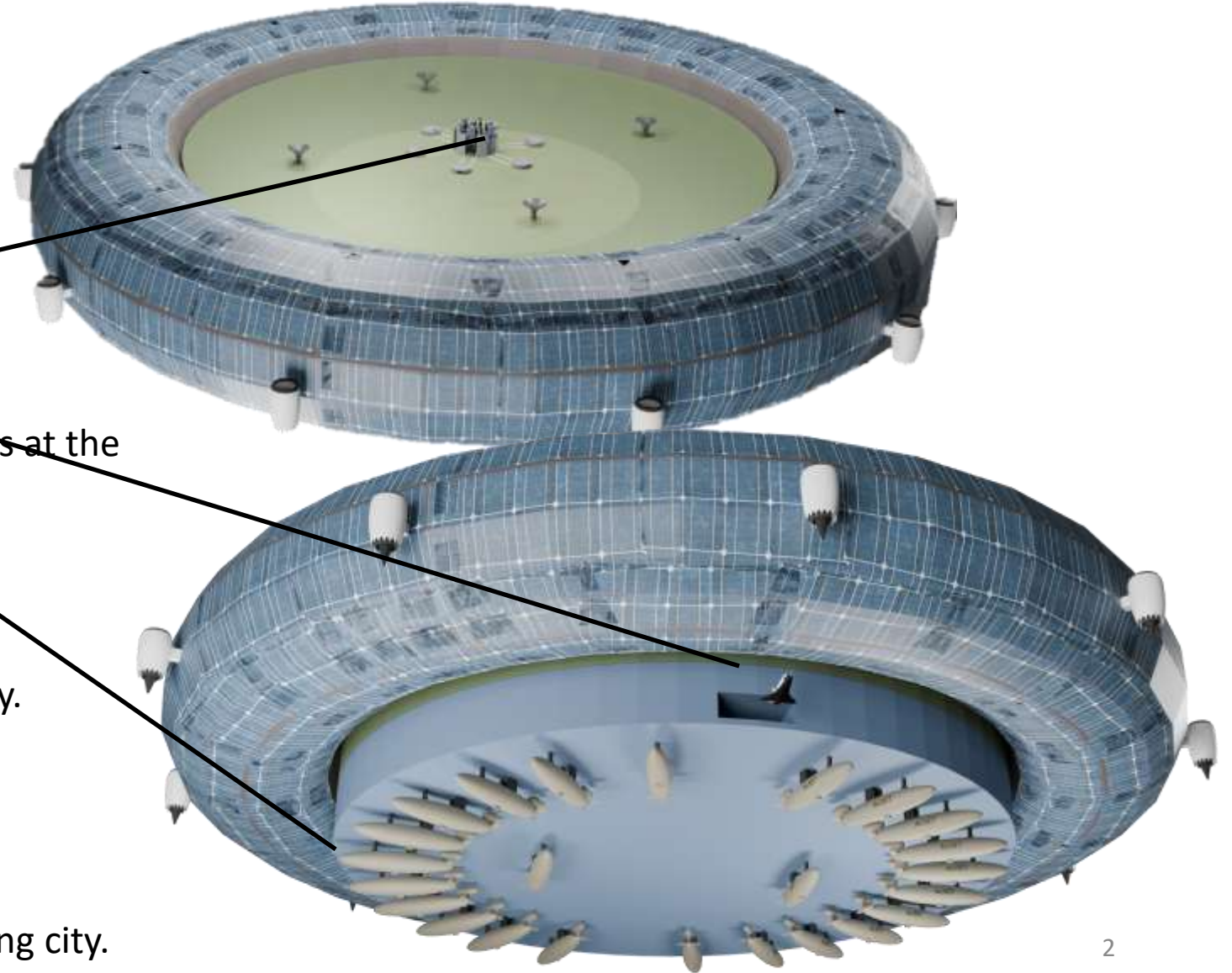
Holding 6 rockets, 6 aerospace
planes and 24 Water Resource Collection Airships at the
same time

Zoning:

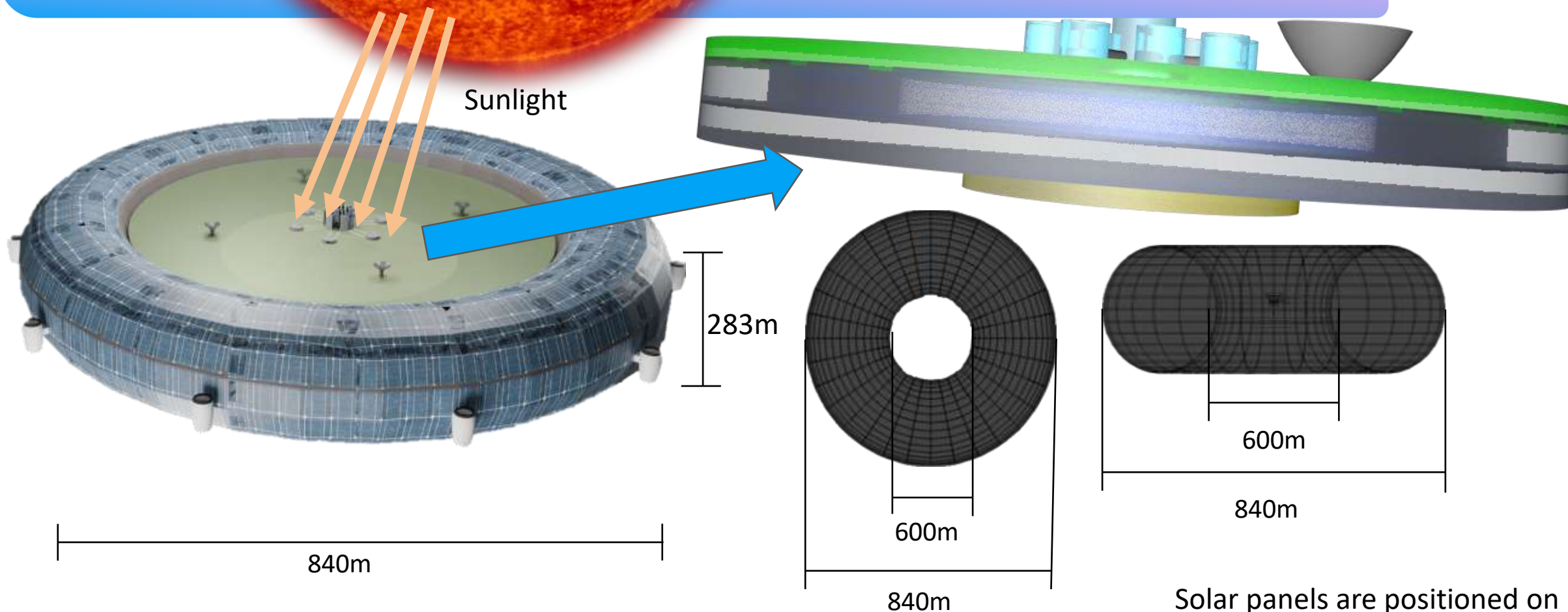
Agricultural area, residential area
and industrial area are integrated in the main city.

Structure:

The main city is surrounded by a
giant balloon with nitrogen inside,
creating buoyancy that can hold the entire floating city.



2.1.1 Overall Design



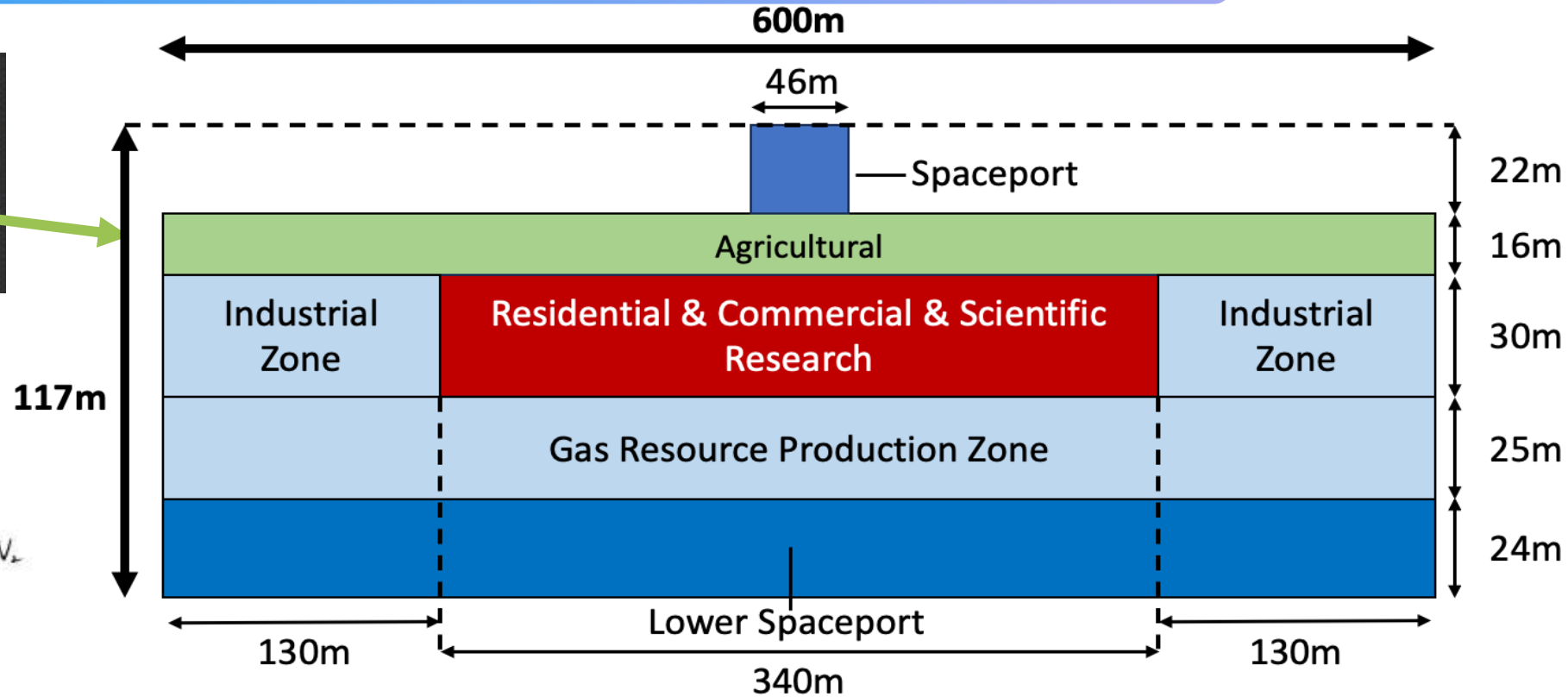
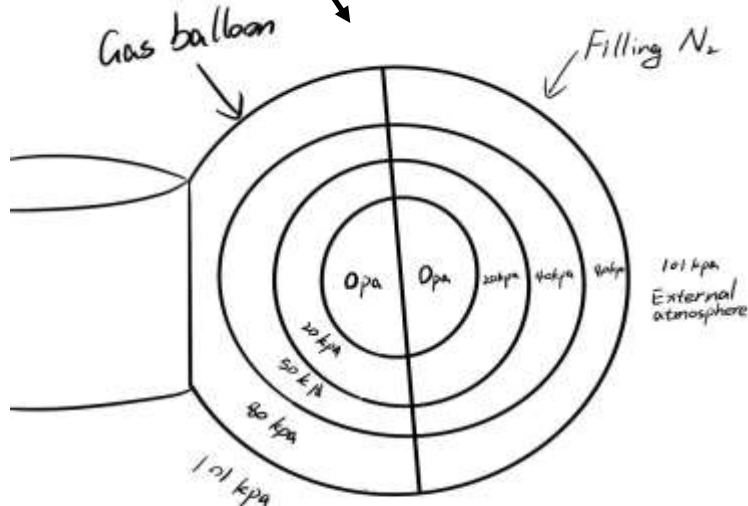
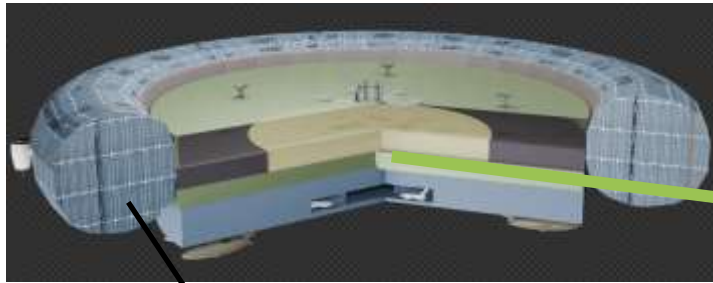
Mass = 6.16×10^6 t

$V_{\text{balloon}} = 7.68 \times 10^6 \text{ m}^3$

Vacancy presents in the sac and an outlay of steel repels the air pressure to maintain its shape and ensure sufficient buoyancy.

Solar panels are positioned on both sides to take advantage of both the reflected light and direct sunlight.

2.1.2 Overall Design



Using partitions, the air bag is divided into many parts to increase the safety of the space city. And nitrogen is injected to reduce the pressure on the airbag material

2.2.1 Construction Sequence

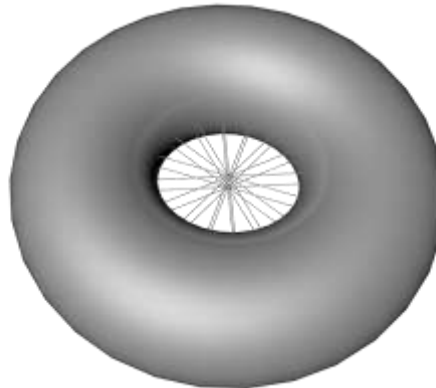
The overall introduction

- With regards to convenience and safety, we decided to

1. Start with stabilizing the general structure
2. Then construct the main sections of the city
3. Eventually the detailed parts.

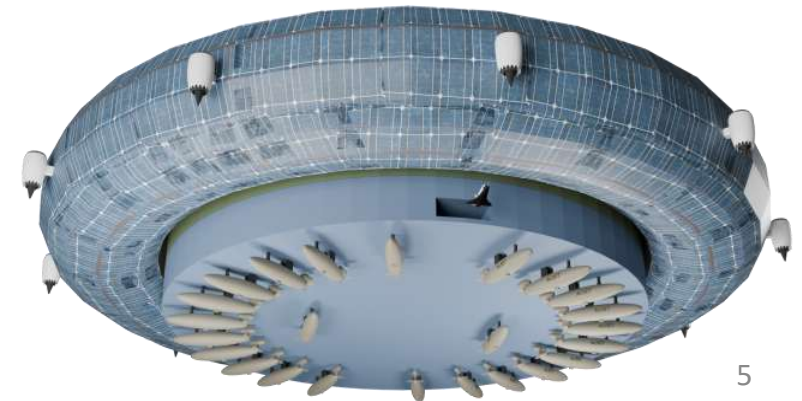
Stage1:

- Set up rigid gas balloons
- Construct Scaffolds and Propellers to stabilize the structure



Stage 2

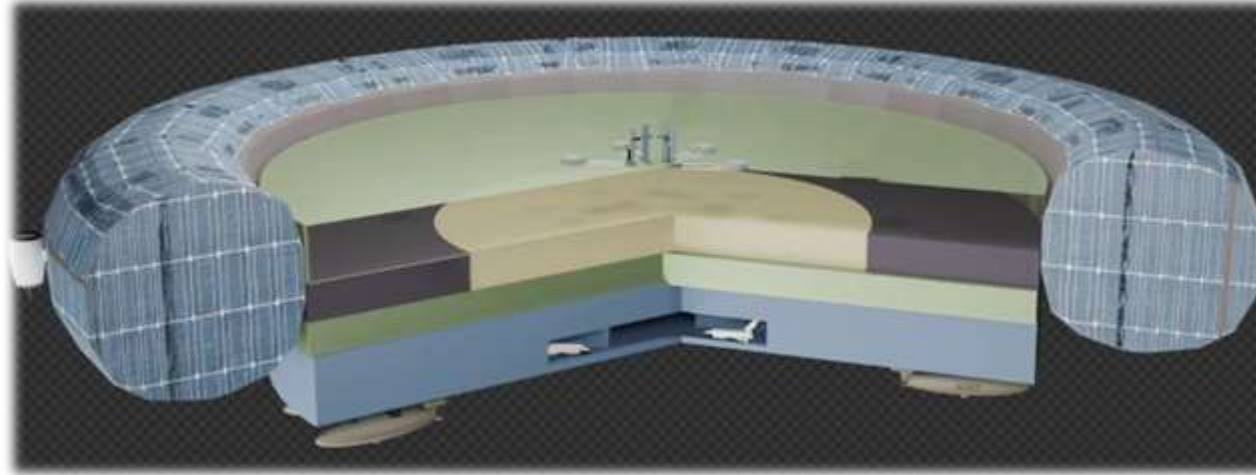
- Build up the upper and downside Spaceports
- Initiate central transportation networks



2.2.2 Construction Sequence

Stage3:

- Build the main bodies
 1. Industrial area
 2. Residential area
 3. Agricultural area



Stage4:

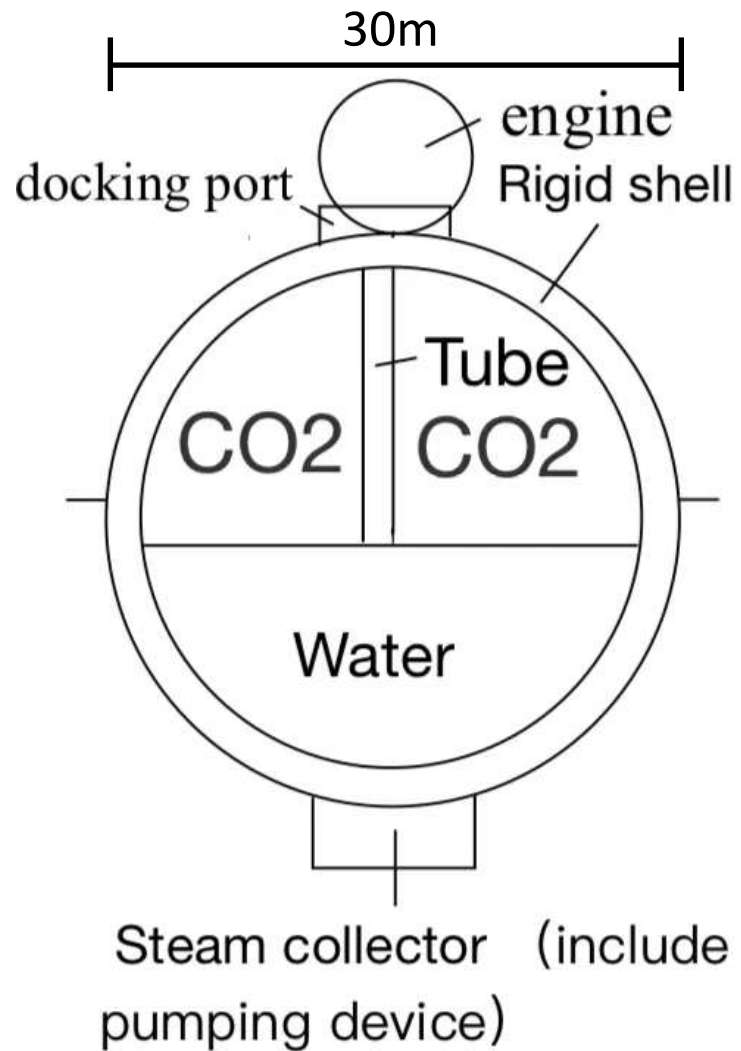
- Develop the complete transportation system
- Attach the solar panels outside the city with the sequence from the inner part to the outside

Stage5:

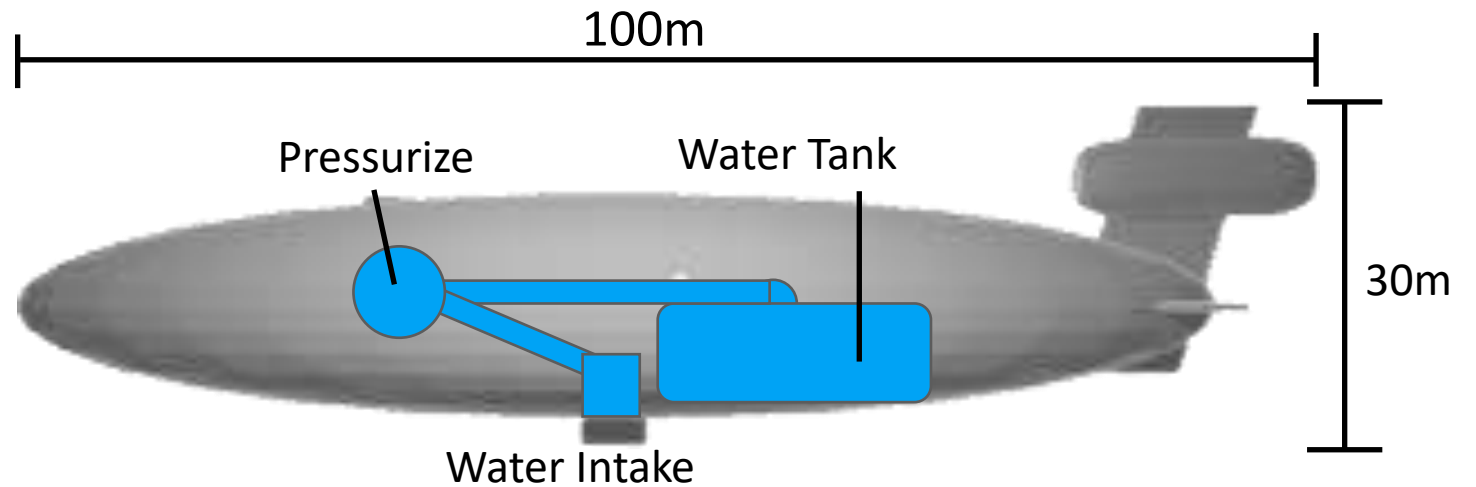
Finally, construct detailed areas like the antennas and set up Internal facilities



2.3.1 Water Resource Collection Airships



(Water-Absorbent Material: **Aerogel**)



Cabin Wall Thickness: 5cm

Length: 100m

Width & Height: 30m

Weight (Empty): 84.47t

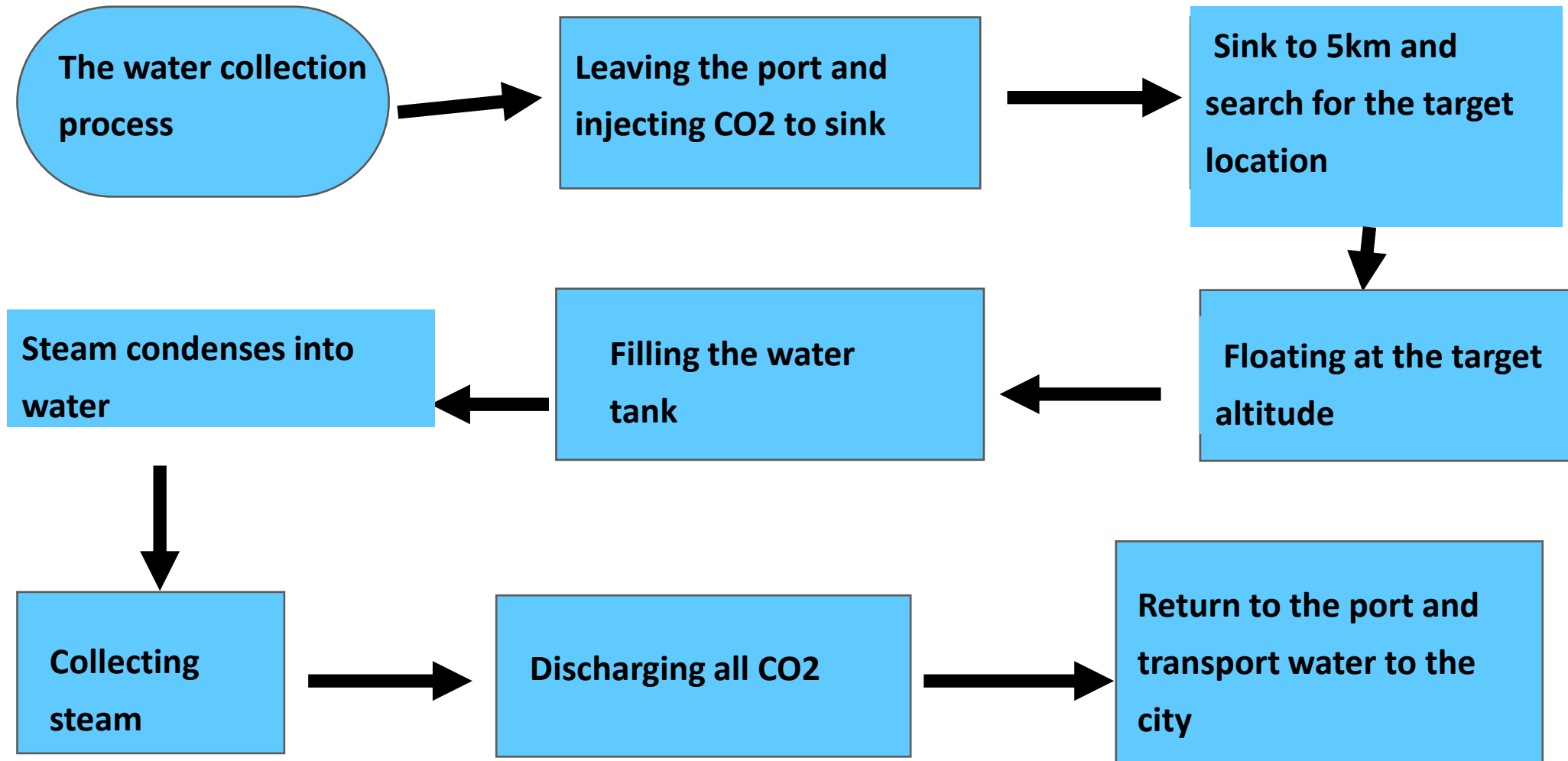
Volume: 70716.8m³

Buoyancy Yield: 134t

Injectable Water Mass: 15.24t

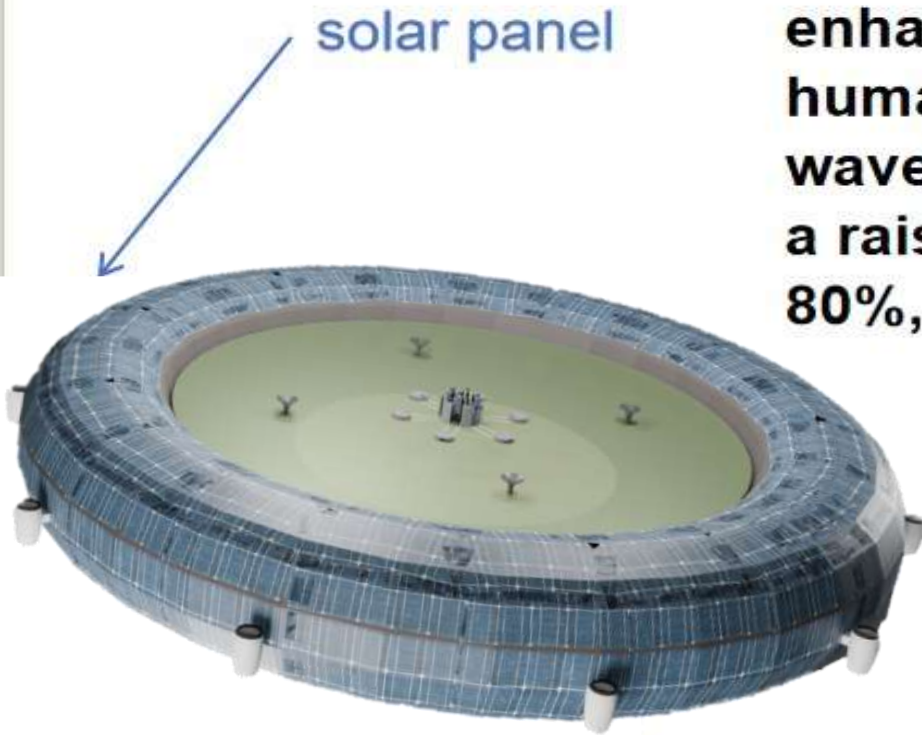
2.3.2 Water Resource Collection Airships

The diagram shows how water is processed after entering the spaceship.



3.1.1 Power supply

The power supply for the city mainly relies on solar panels on the rigid air bags. With the development of nanotechnology and enhanced spectrum sciences, which enables humans to utilize lights with a wider range of wavelength. Hence, we believe that there will be a raise the efficiency of the panels from 35% to 80%, in the future 40 years.



As the refractive index for the Venus atmosphere reaches 96% Auset is exposed to all-direction sunlight, it is necessary to cover the whole city with solar pannels, even for the backs. This action can significantly raise the efficiency of space energy usage.

3.1.2 Power supply

STATISTICS	PROJECT	
$P/S=2613.5W/m^2$	Largest power	762723kW
$C=3642.4m$	Energy out put	18305372kWh
$x=283m$	per Industry	700000kW
$S=380000m^2$	power	
Efficiency=80%*96%	Residential reserve power	3100kW
	Agricultural needs	41000kW
		Highest power consumption a day 17913600 Kwh

In the city, we mainly use stacked solid state batteries which are efficient and reliable enough.

Solid-State Lithium-Metal Batteries



EMERGENCY MINIMUM	LIVING	ENERGY	NEEDS
Industry to maintain life		350000kW	
Residential needs.		2800kW	
Agricultural		26000kW	
TOTAL COST IN 60 DAYS. $(350000 + 2800 + 26000 + 2300)kW \times 24 \times 60 = 548,784,000 kWh (60days)$			

3.2.1 Water Cycle

Key advantages of the cycle process:

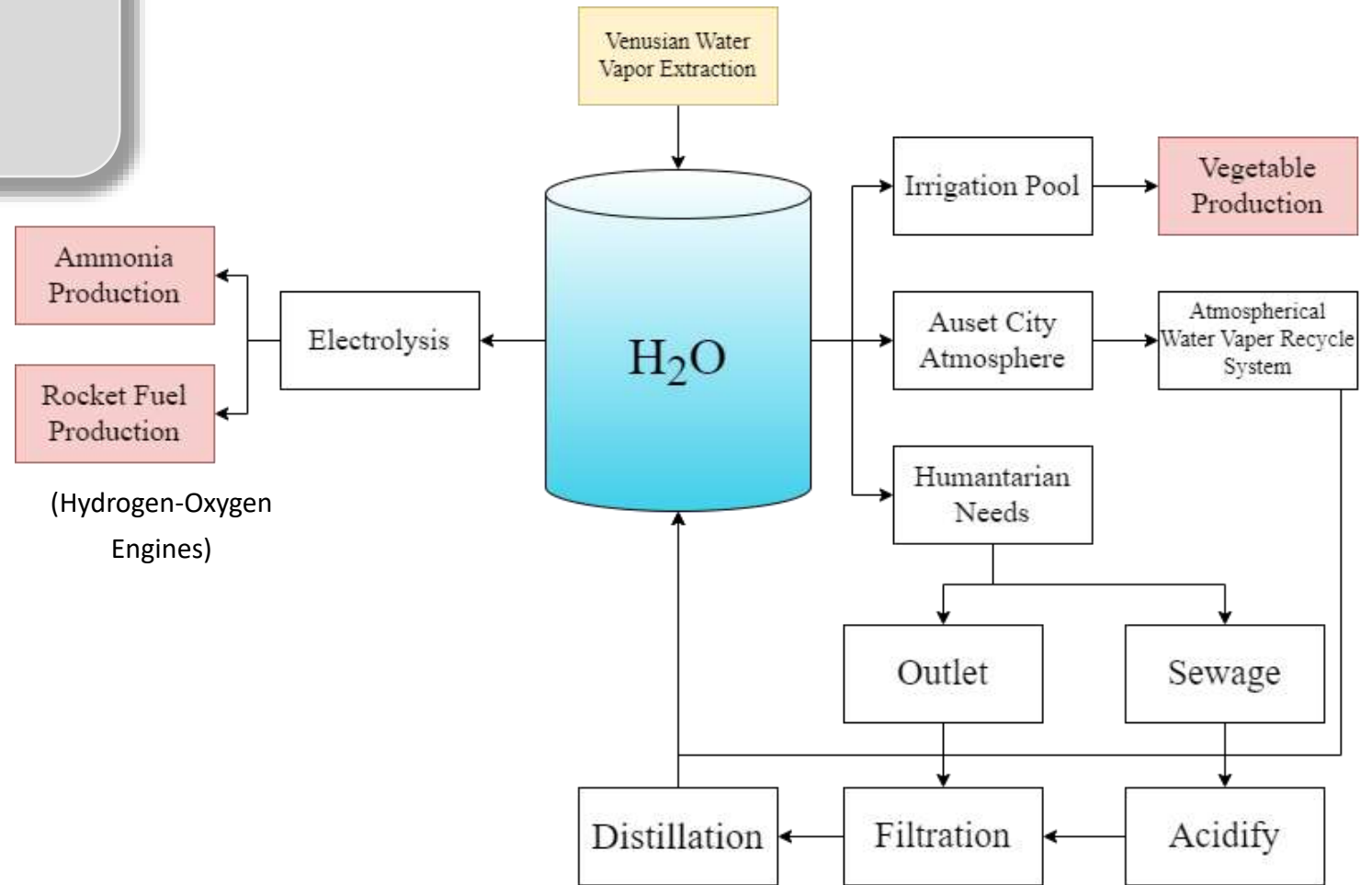
High economy;
Dynamic equilibrium;
Self-sustain in an emergency.

Water supply includes:

-Water vapor gathered from Venusian atmosphere (~2800 t)

Water output includes:

-Rocket fuel production(~2400 t)
-Agricultural products(~400 t)



3.2.2 Gas Cycle

Carbon dioxide supply:

- Carbon dioxide gathered from Venusian atmosphere (~600 t)

Carbon dioxide output:

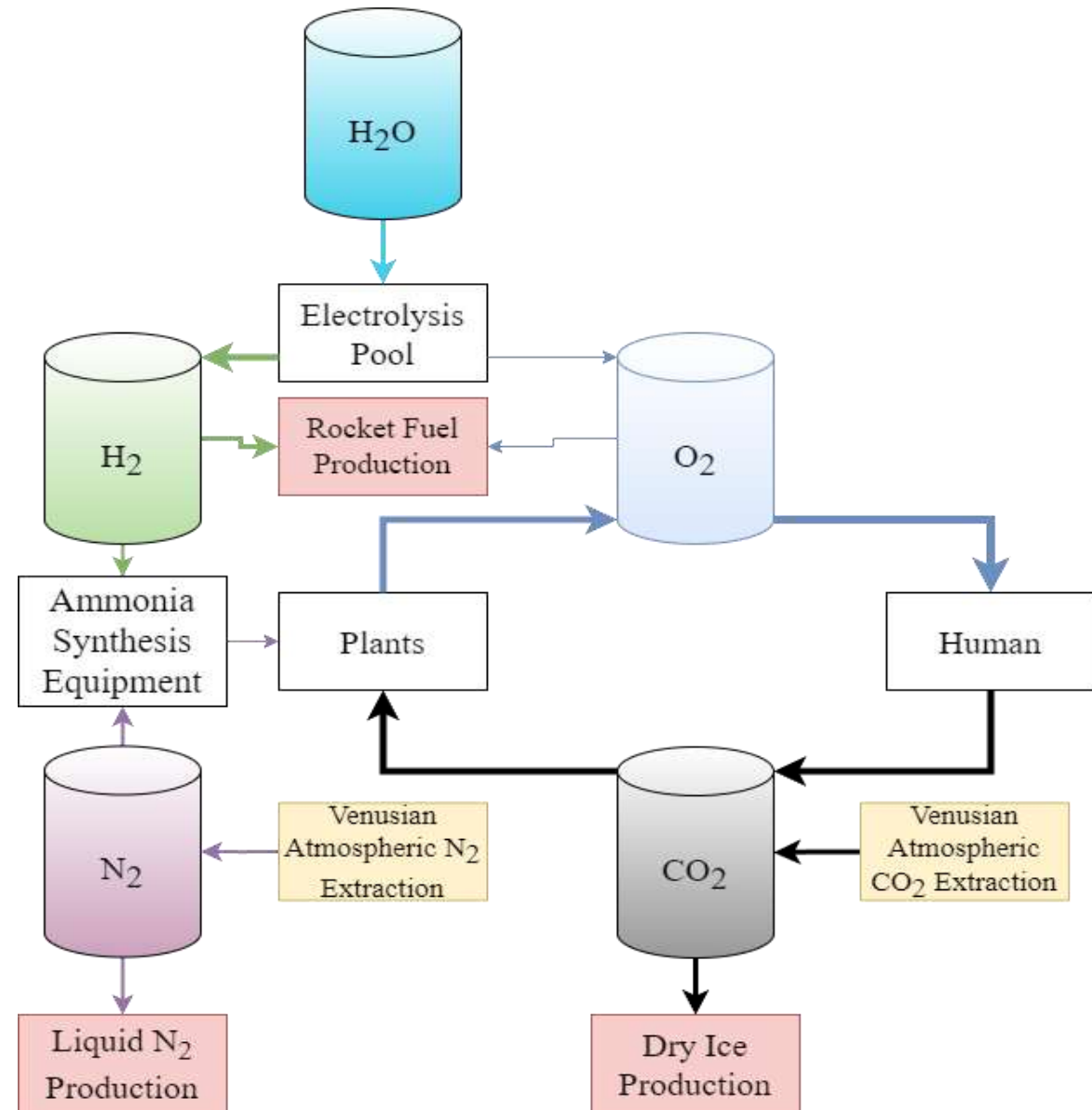
- Dry ice export (~500 t)
- Chemical Industry

Nitrogen supply:

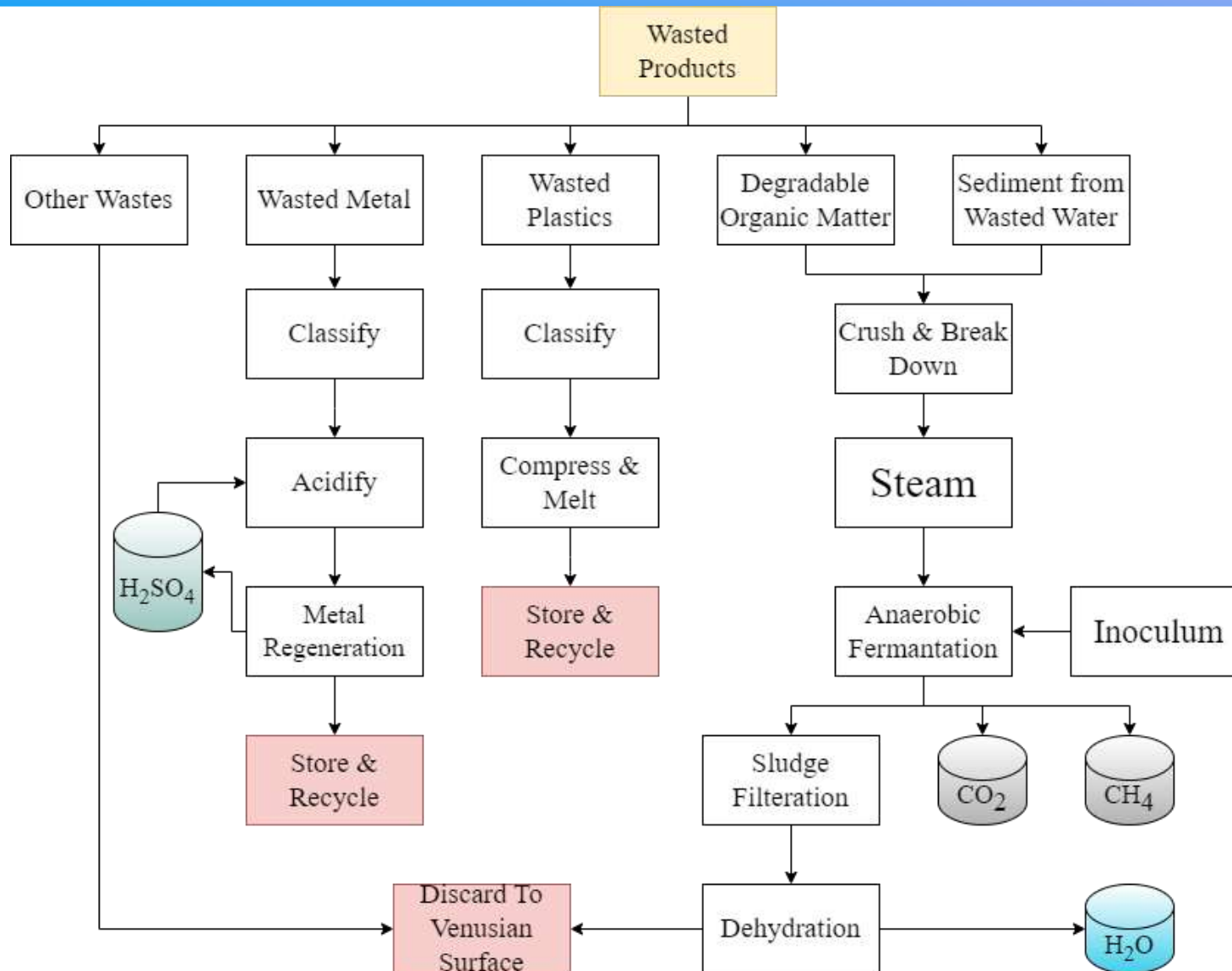
- Nitrogen gas gathered in upper Venusian atmosphere

Nitrogen usage:

- Produce liquid nitrogen for export
- Produce ammonium fertilizer for agriculture



3.2.3 Waste Processing



Waste processing system input:

-Waste materials and sewage sediments

System output:

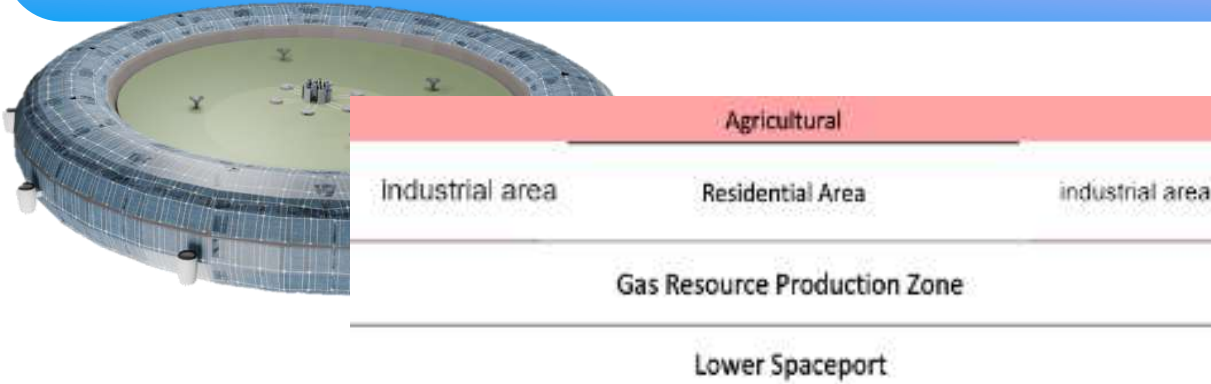
-Recycled useful gases of CO_2 and H_2O

-Metal sulfates

-Simple organic compounds

-Unwanted residue

3.3.1 Food Production



- A stereoscopic design-----space saving
- Mixed cultivation methods----enhanced productivity
- Artificial automaton assistance---economical
- Multi-functional usage of gas-----efficient operation
-

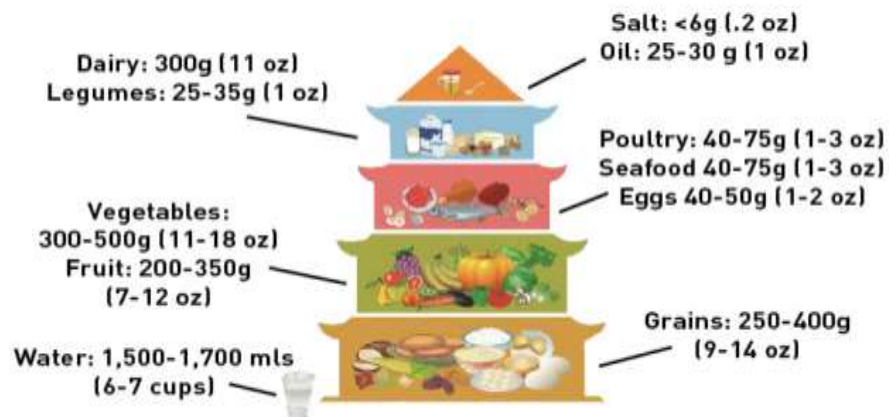
Approximately 513 tons of assorted ingredients are produced everyday. By using an alternating agricultural design, this section can ensure sufficient food supply for the whole city.

Type	Area (hectares)	Growth Cycle (days)	Cultivation Method	Average Daily Yield per Hectare (tons)	Average Daily Yield (tons)
Tomato	20.5	90	Gas Cultivation	3	60.5
Cabbage	20.5	40	Gas Cultivation	5.1	104.55
Watermelon	23	90	Gas Cultivation	1.8	41.4
Pepper	19	100	Gas Cultivation	0.6	11.4
Lettuce	30	80	Gas Cultivation	0.625	18.75
Onion	25	100	Gas Cultivation	0.75	18.75
Garlic	11.5	120	Gas Cultivation	2.025	23.29
Eggplant	27.5	180	Gas Cultivation	0.42	11.55
Soybean	49.8	90	Soil Cultivation	1.8	89.64
Potato	61.8	90	Soil Cultivation	1.67	103.21
Wheat	59.8	180	Soil Cultivation	0.5	29.9
Total	348.4	-	-	-	512.94

3.3.2 Food Production

	Energy	Proteins	Carbohydrates	Fat	Microelements and Vitamins
nutrition for 3500 people	8750000 Kcal	2800 g	21000 g	4200 g	8050 g
types of food	the mixture of them	soybeans and its processed food	potatoes and wheat	soybeans and its processed food	all fruit and vegetables
approximate intake quantity	875kg	7.778 kg	124 kg	262.5 kg	2975 kg

China's 2016 recommended daily dietary intake

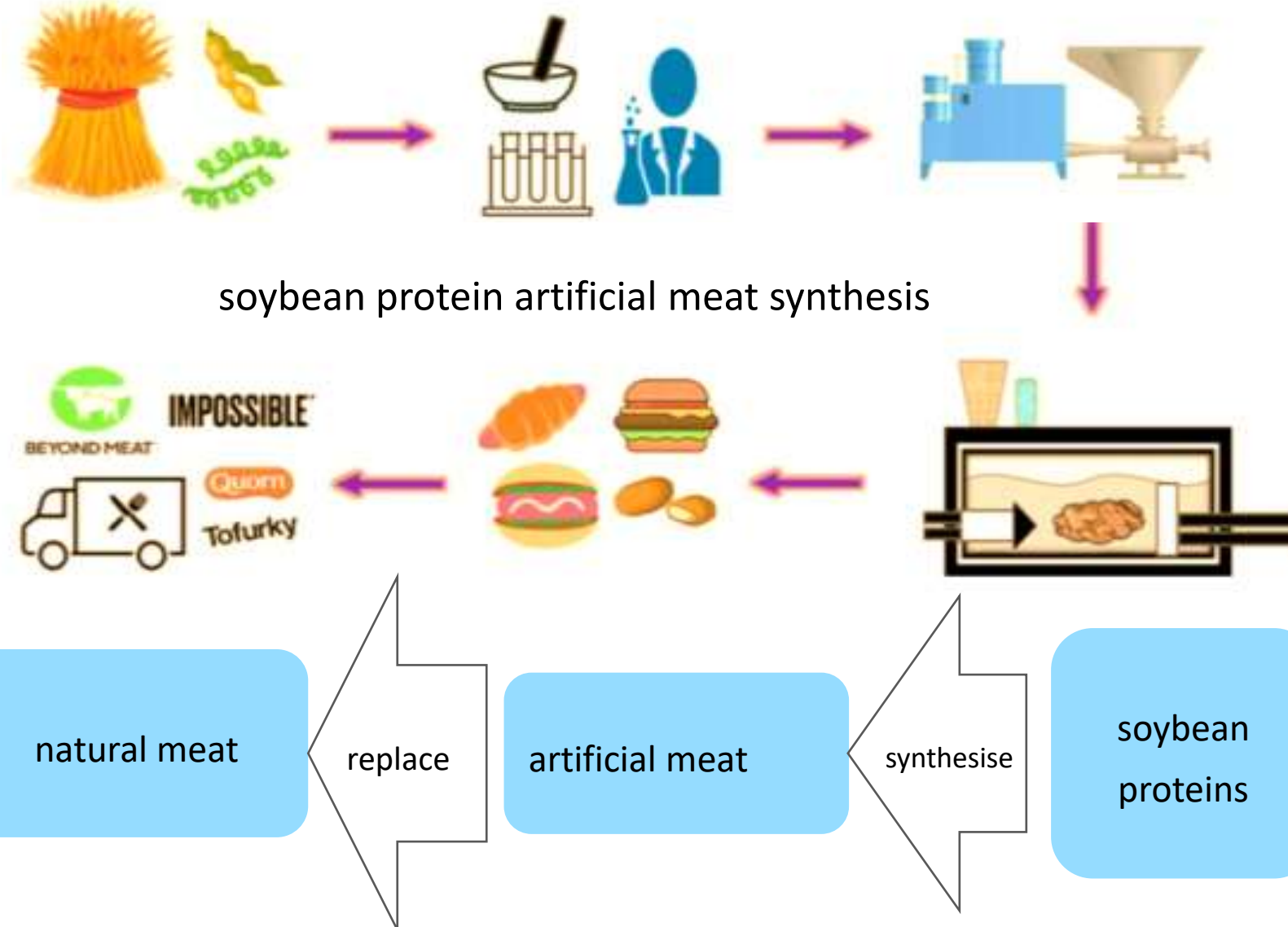


Source: Chinese Nutrition Society

Over 10 kinds of food provide residents with various choices and nutrition assurance.

This chart takes the dietary pagoda into account and it indicates the ability of our food system to fulfill all the human needs.

3.3.3 Food Production

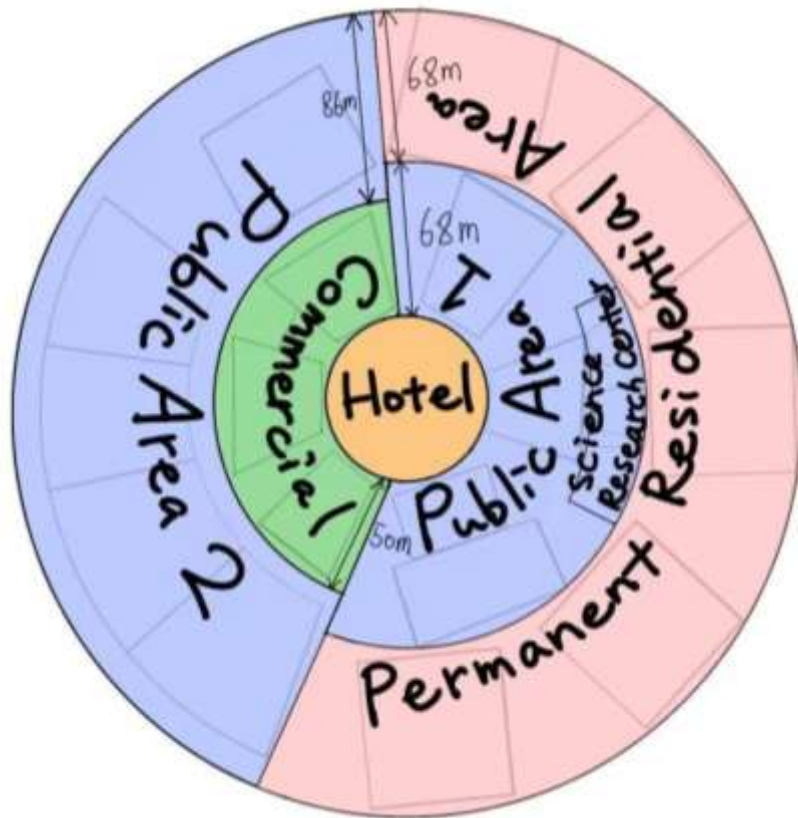


Due to the high cost to develop grazery, we innovatively decide to take advantage of proteins from soybeans to replace natural meat.

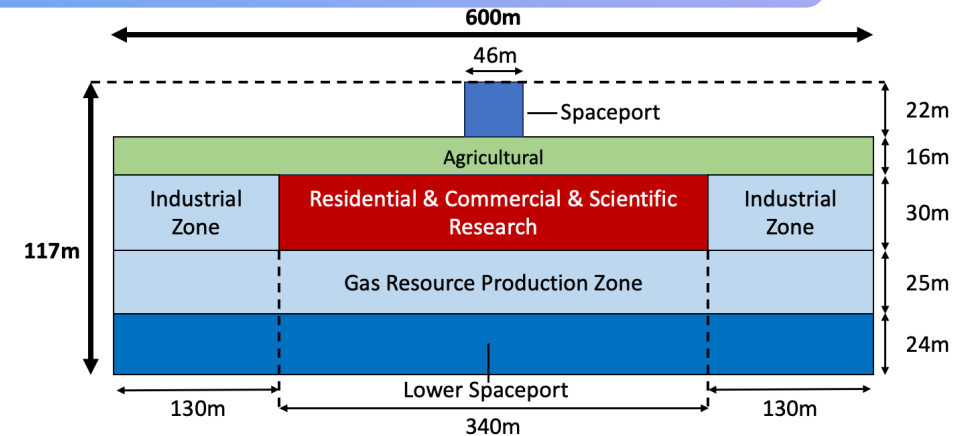
This illustration represents a general process by which soybean proteins are synthesised to artificial meat.

4.1.1 Residential Area and Public Spaces

Location of the Human Factor Area



Distribution plan of the Human Factor Area

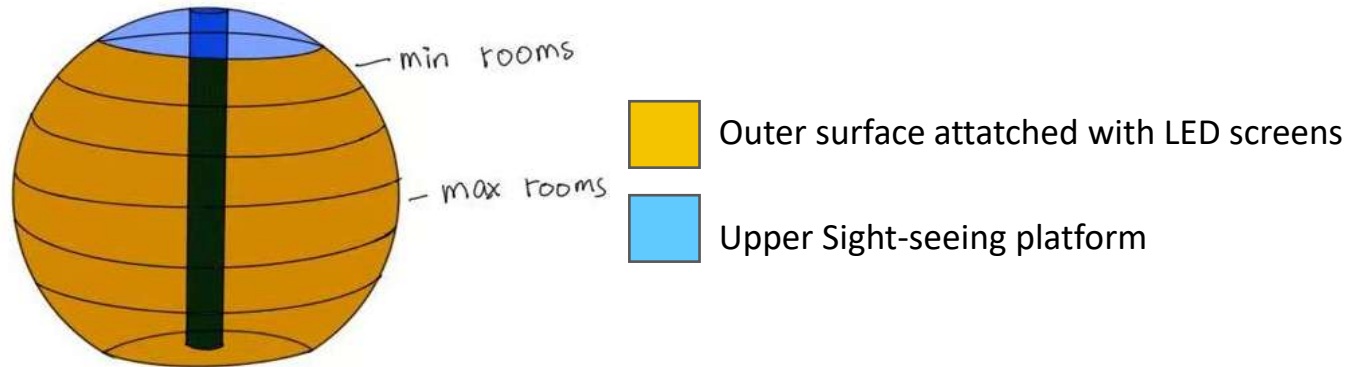


Location advantages:

- The Auset Space Hotel: located in the middle for convenient travel of visitors.
- The Permanent residential area: connects with the two public areas for easy access to relaxation and recreational activities and the science research center.
- The commercial area: set near the center of the whole area which makes all residents easy to access.
- The two public areas: smaller area for daily necessities, larger area for entertainment activities.

4.1.2 Residential Area and Public Spaces

Auset Space Hotel



The Auset Space Hotel:

- constructed in the center of the Human Factor area
- consists of **7 floors and approximately 250 rooms.**

LED screens: **present the outer appearance of Venus**, giving guests a sense of living on Venus, providing **immersive experiences.**

Upper Sight-seeing platform: provide guests with a **full scene of the whole Human Factor Area**, giving them a **deep impression.**



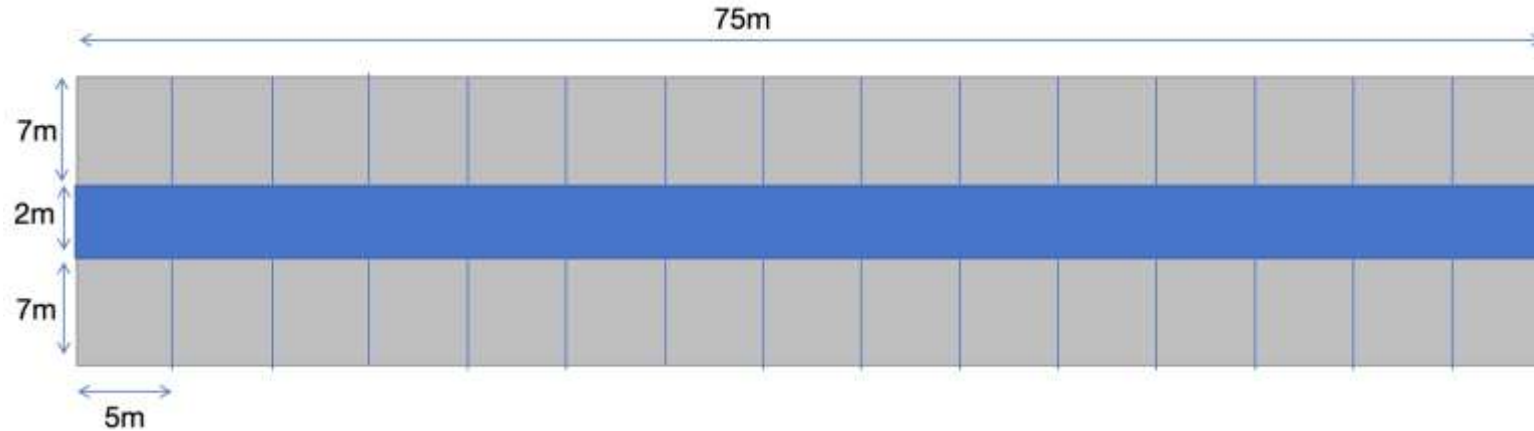
Total height: 24 m

Radius: 12.5 m

Total volume: 8181 m³

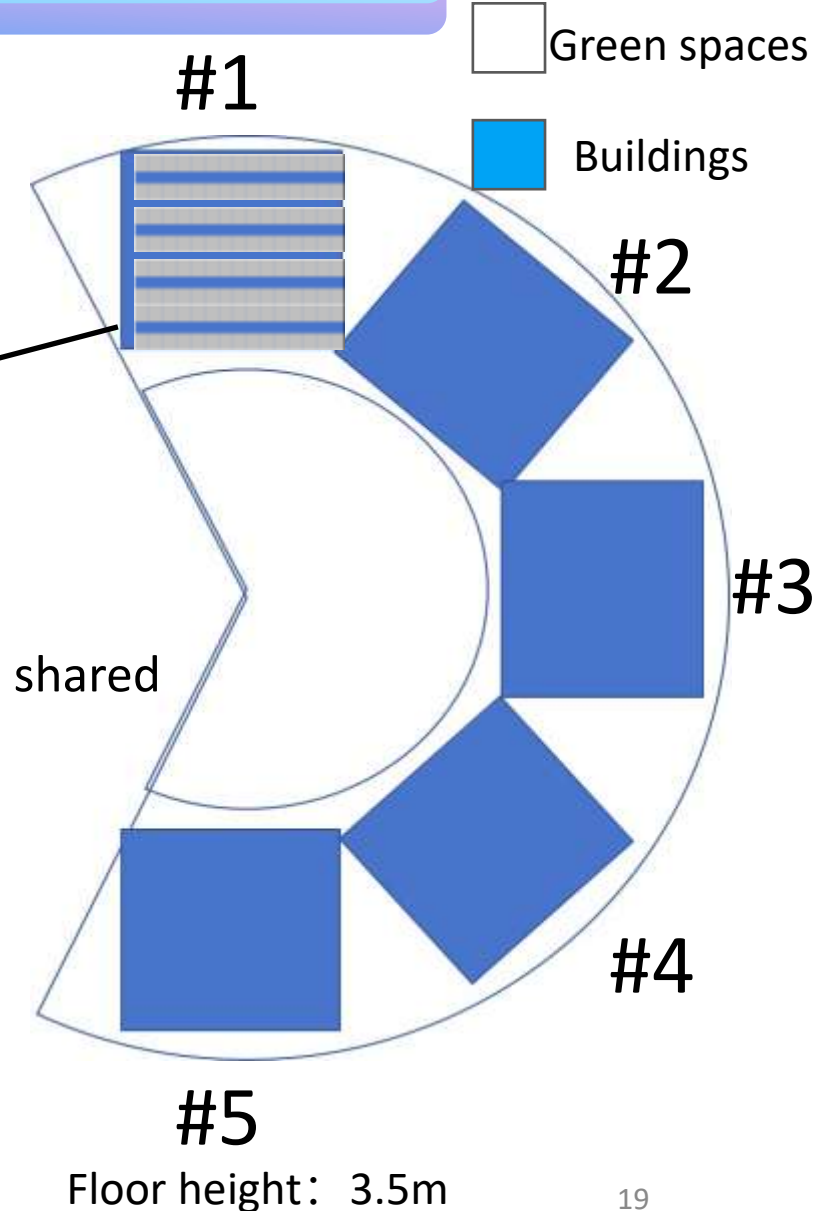
4.1.3 Residential Area and Public Spaces

Permanent Residential Buildings



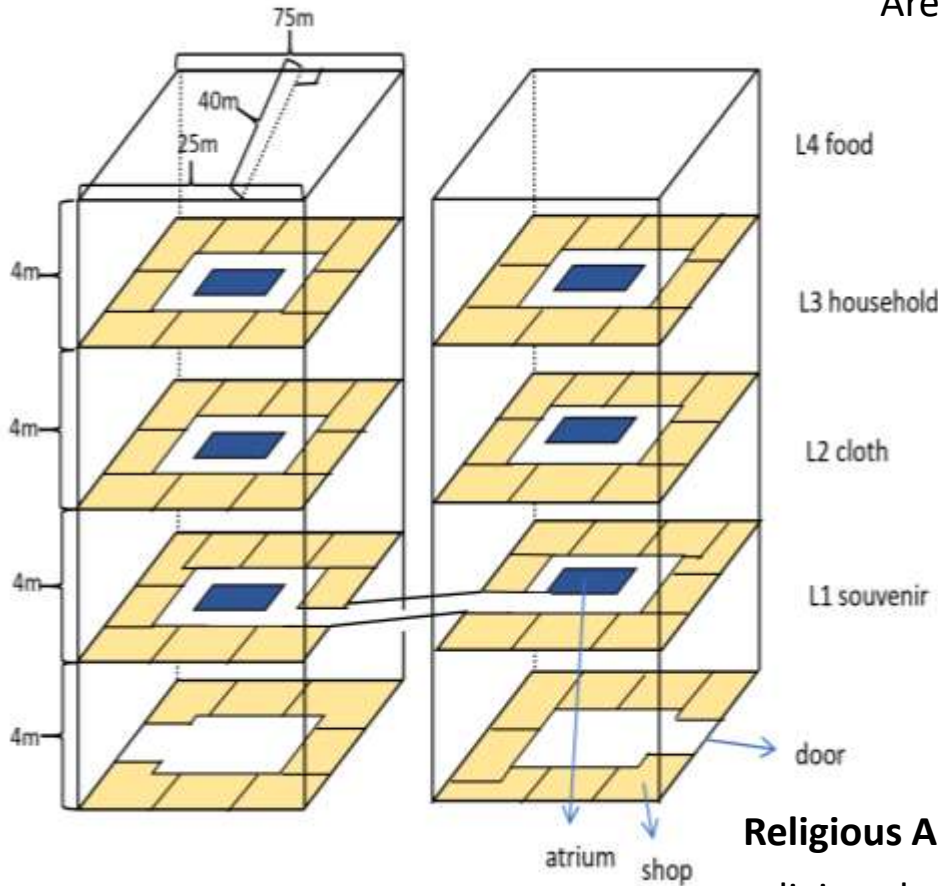
Top view of one section

- 5 Buildings, each with 6 floors and 4 The buildings will consist of different shared spaces:
- For each building, a special shared place will be located in the third floor.
- Building 1 Gymnasium
- Building 2 Shared Kitchen
- Building 3 Bar
- Building 4 Club Space
- Building 5 Children Space



4.1.4 Residential Area and Public Spaces

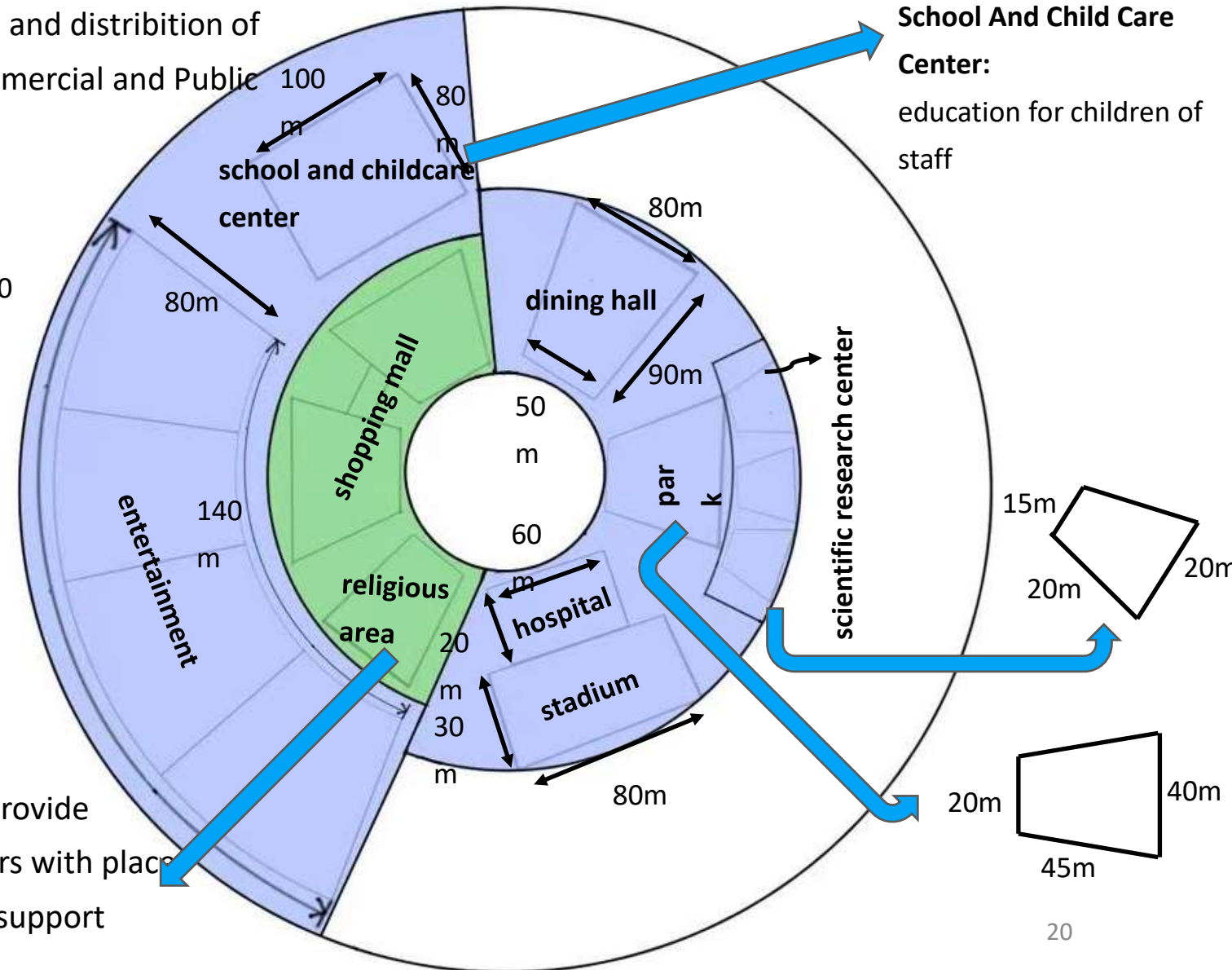
Commercial and Public Areas



Floor plan of the Commercial Halls

Location and distribution of the Commercial and Public Area

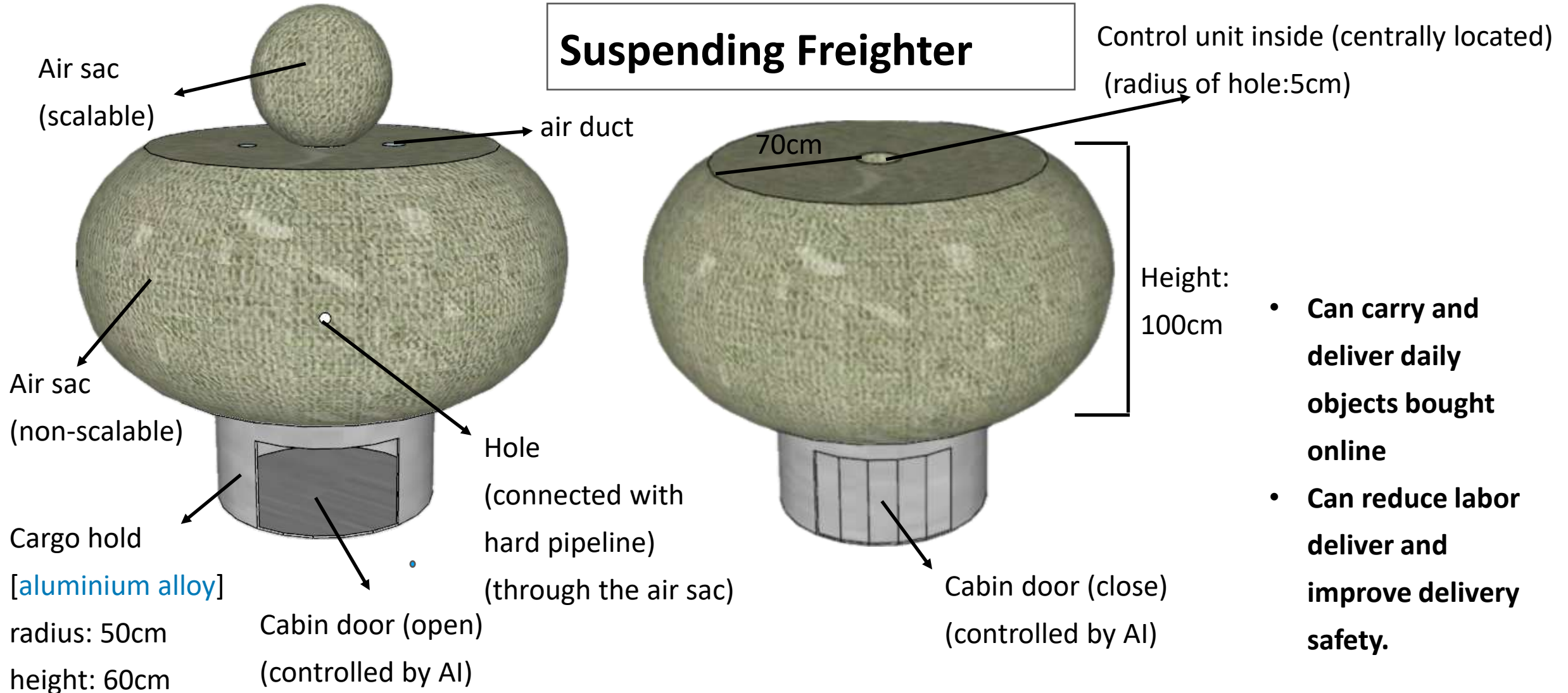
Religious Area provide religious believers with place to gain spiritual support



School And Child Care Center:
education for children of staff

4.2.1 Automation Devices

Suspending Freighter

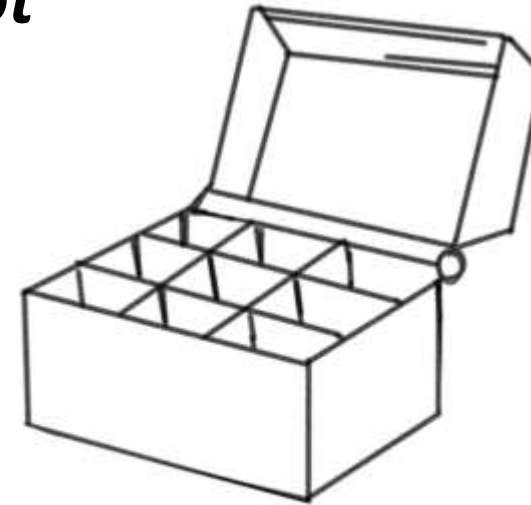
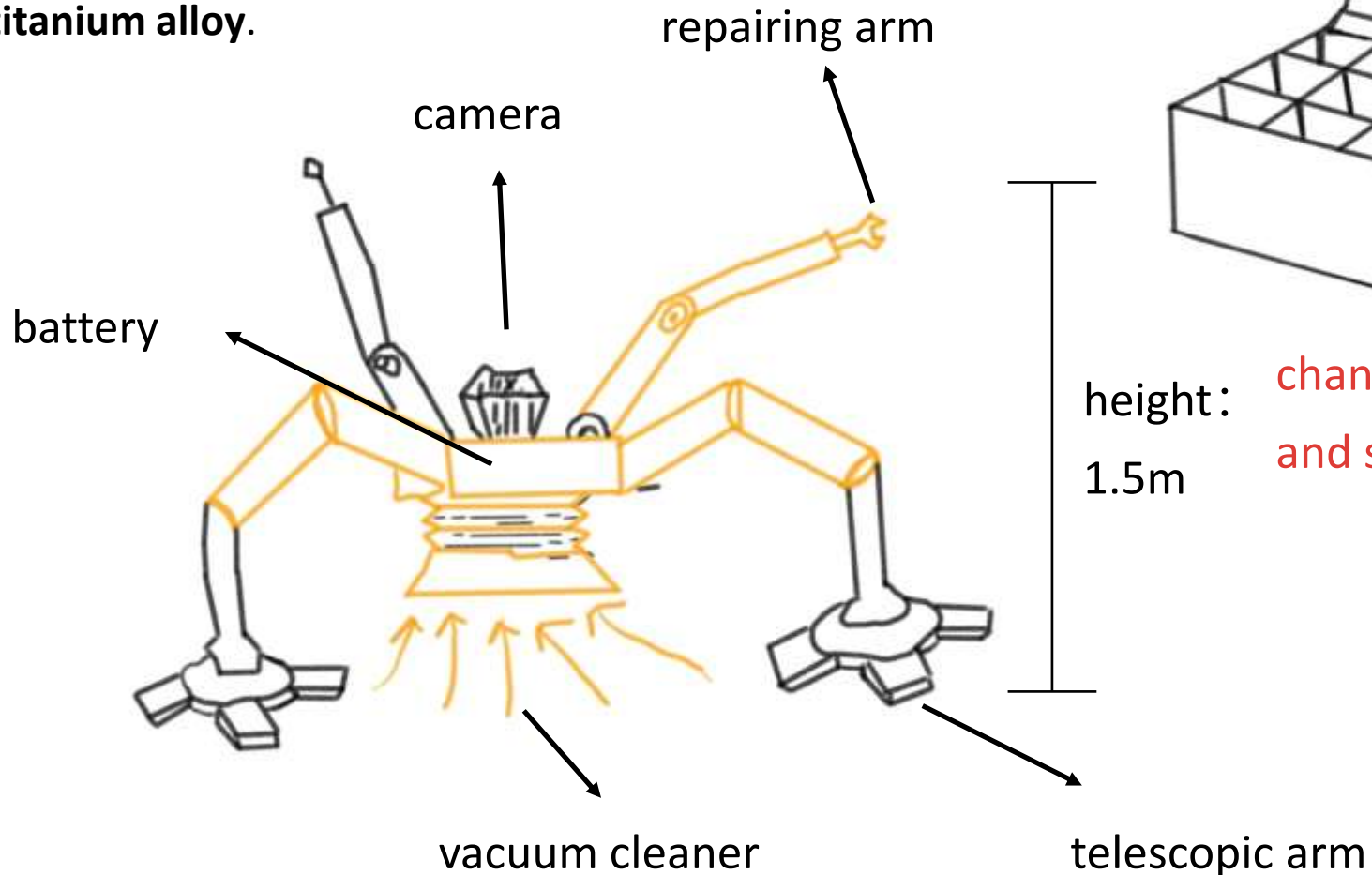


- Can carry and deliver daily objects bought online
- Can reduce labor deliver and improve delivery safety.

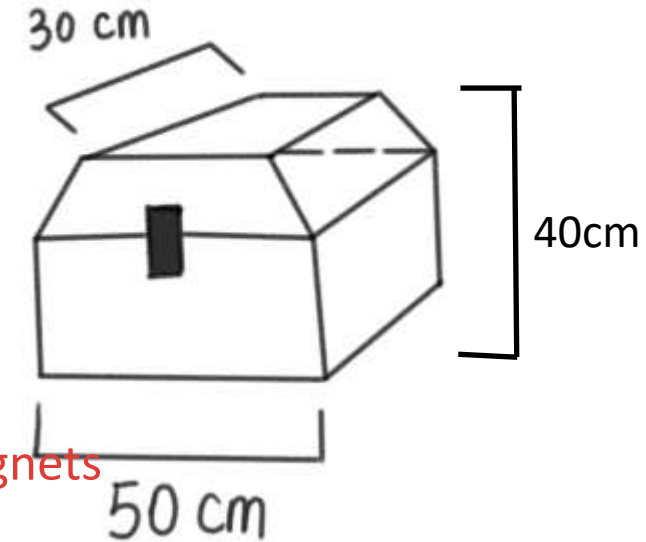
4.2.2 Automation Devices

intelligent repair bot

The material of the entire machine is **titanium alloy**.



Tool Box



change the tools: magnets
and snaps.

Advantages:

- Fix precise items and save human labor
- Reduce the cost of broken items in the long-term

4.3.1 Entertainment and Recreation Activities

Realistic Room Escape

900 square meters with LED screens on walls

50 players / 2 hours

safety prioritized

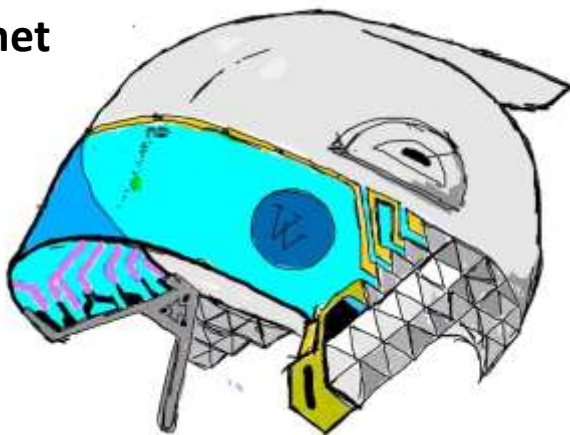
sense of cooperation

combining education with entertainment

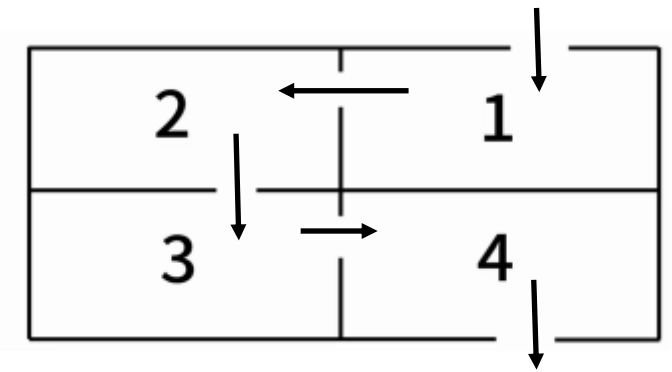
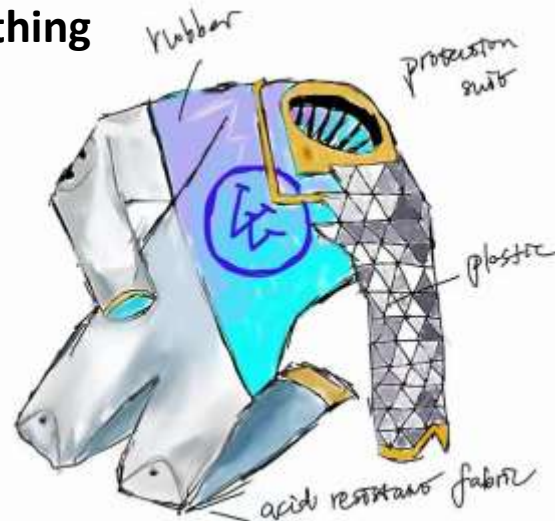


LED scenario

Helmet



Protective clothing



Room distribution

4.3.2 Entertainment and Recreation Activities

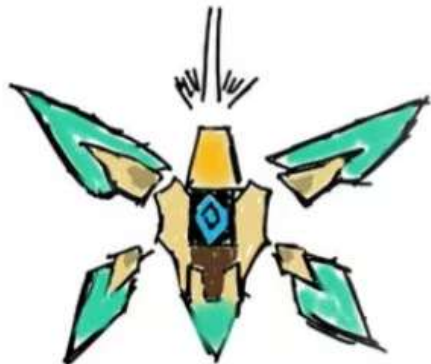
The Eye of Venus

- VR Exploration Program

Integrating space technology, VR, and remote control technology
control system and automatic navigation function

preset routes

"airplane version bumper car"



This device is used for real exploration on Venus which gives real-time video and audio feedback to the player.

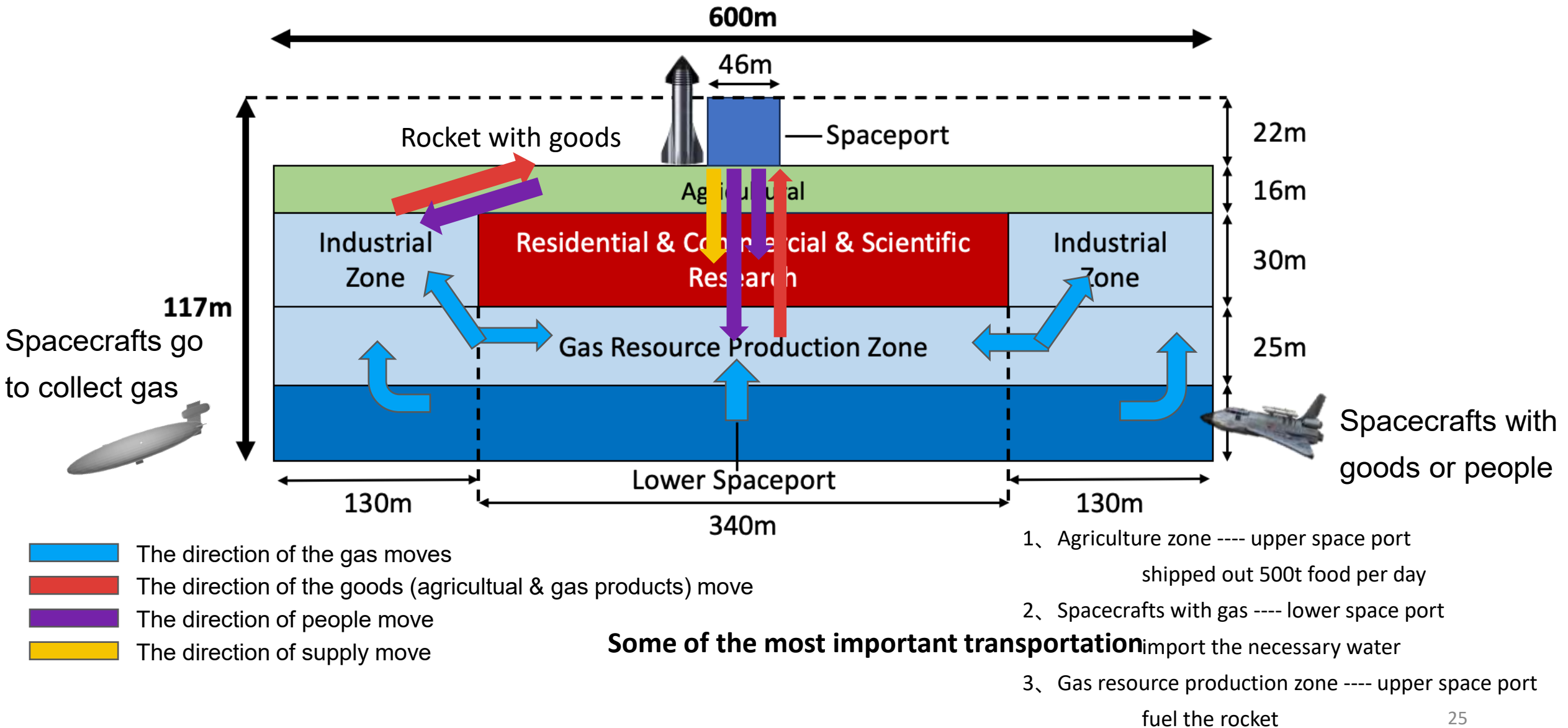
Flying environments



- Application of the Venusian environment
- Can attract those who are interested in explorations but don't want to take risks

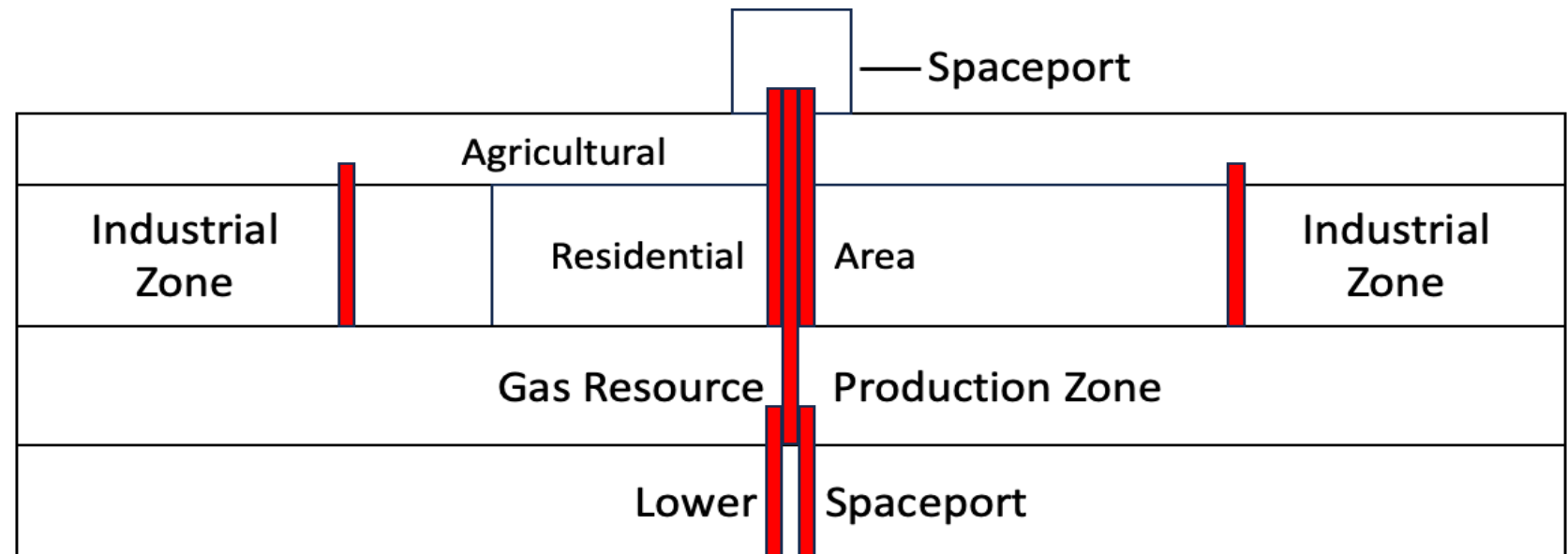
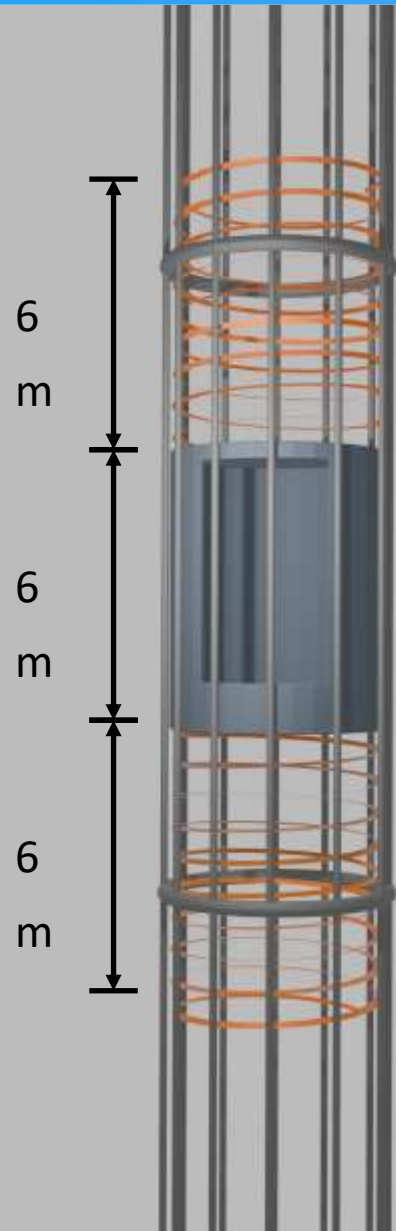
5.1.1 Transportation

25



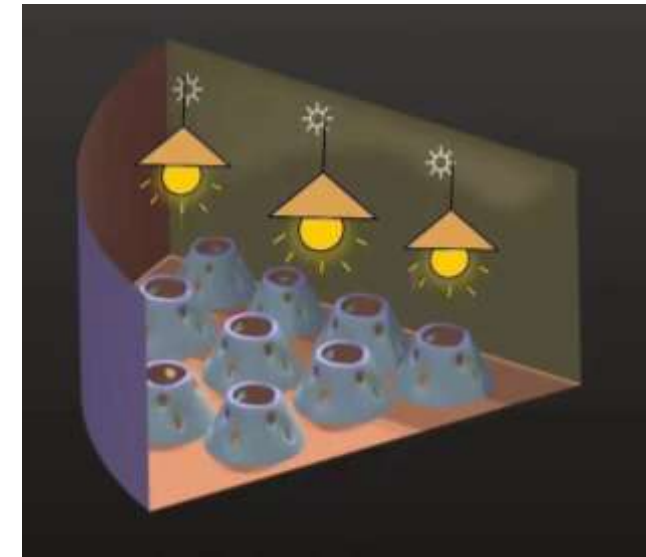
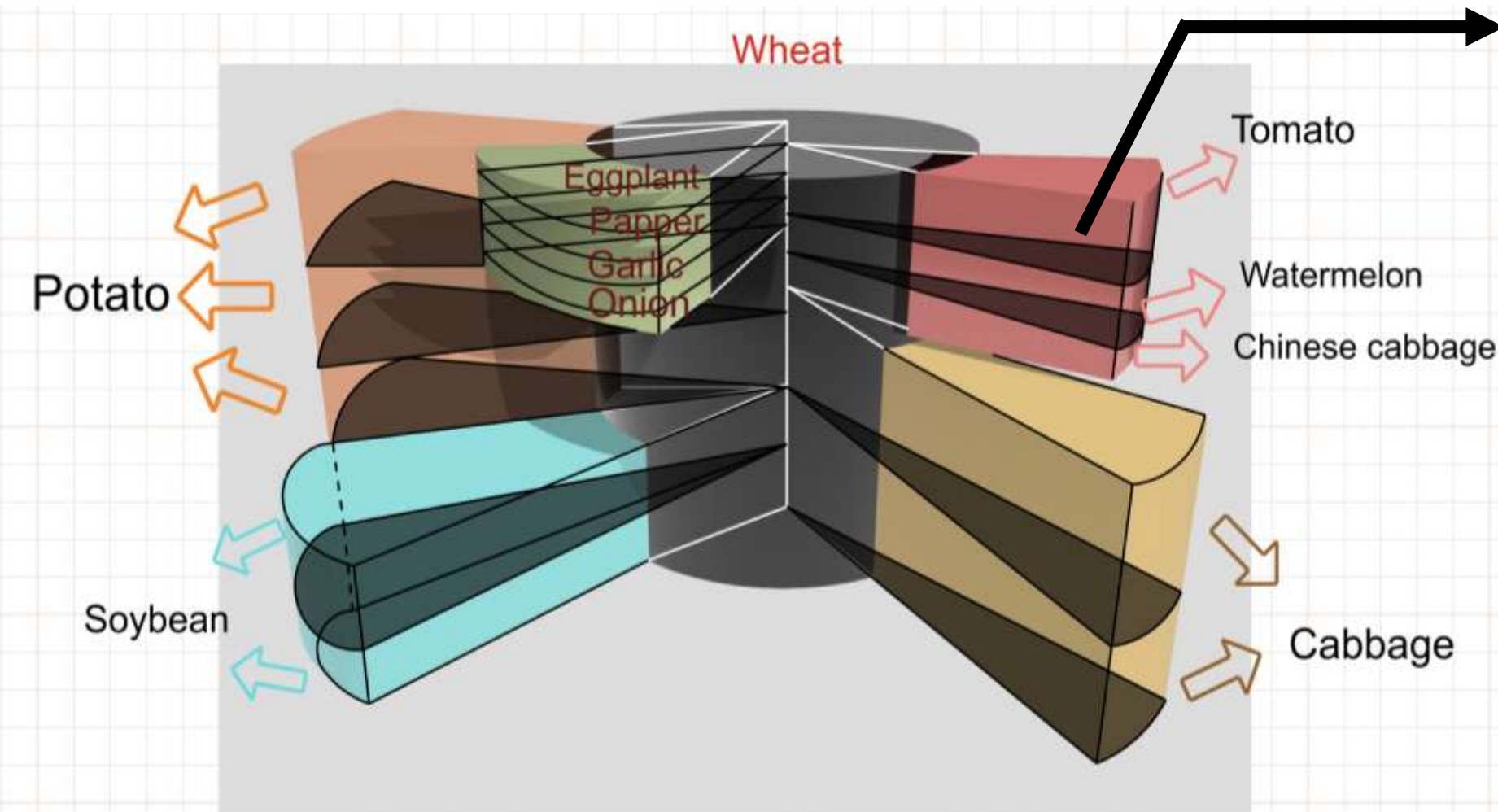
Electromagnetic Elevator

- Electromagnetic Energy Source
- Operate at a speed of 600km/h through
- Exclusively for solid goods and people transference
- Each area is accessible via elevators



5.2.1 Agricultural Area

Nebulization system



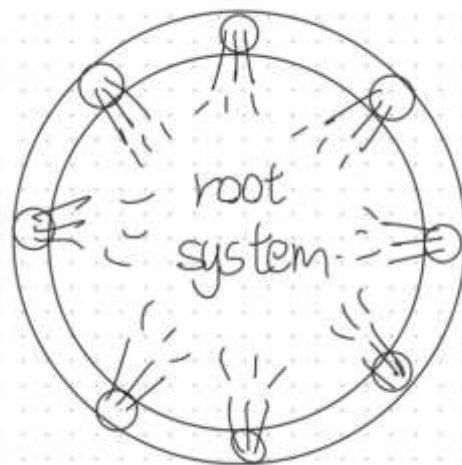
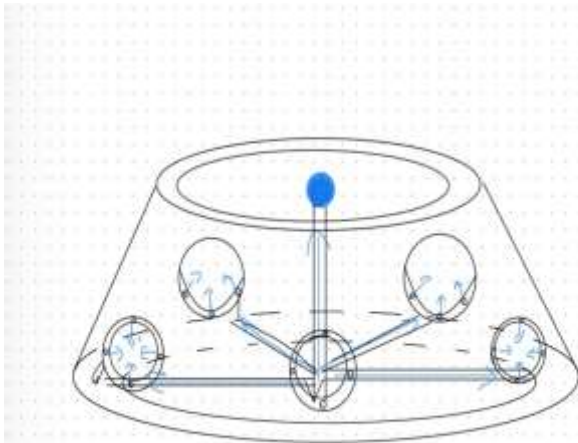
Fog Nutrient Date

*nutrient fog contain Mg Fe N
P microelement

5.2.2 Agricultural Area

Fog cultivation system

Nutrient water reaches each plant from the central water pipe along the wall water pipe and is sprayed in a directional manner in the form of fog.



5.2.3 Agricultural Area

Robot buddy

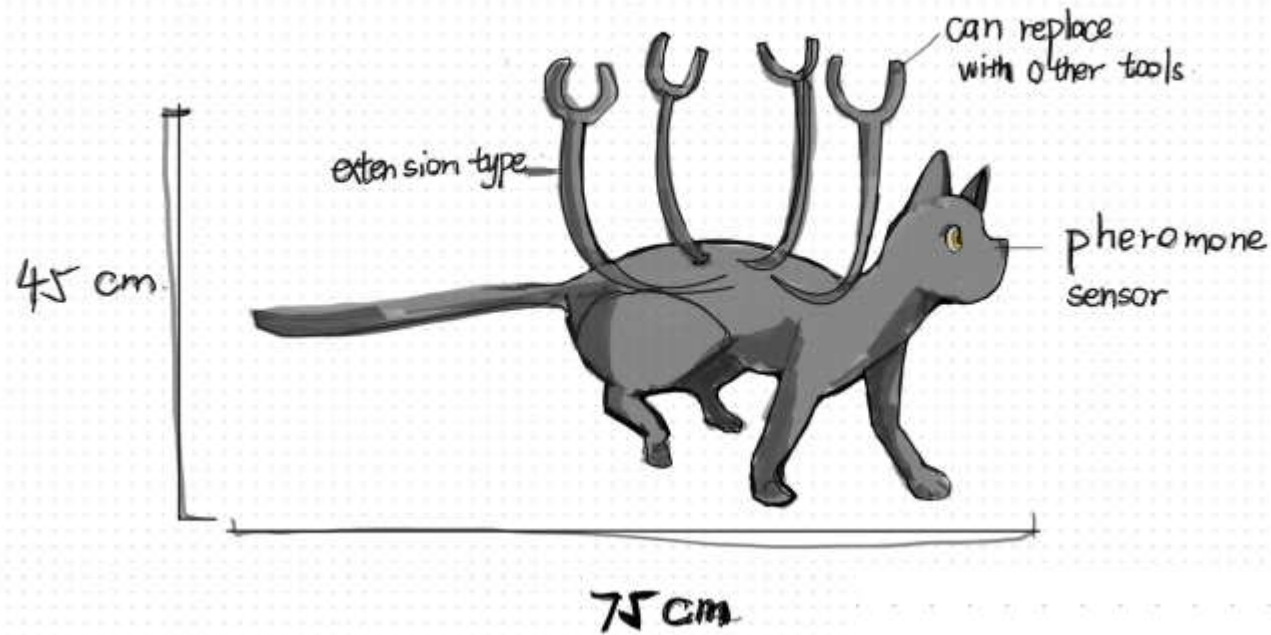
Wonder cat and Hercules

*pertinence

*high sensitivity

*multi-functional

Wonder cat texture: Titanium and
aluminum alloys



Hercules texture: Titanium alloys
aluminum alloys and rubber
Function

Wonder cat: Identify, pick, repair

Hercules texture: Transportation

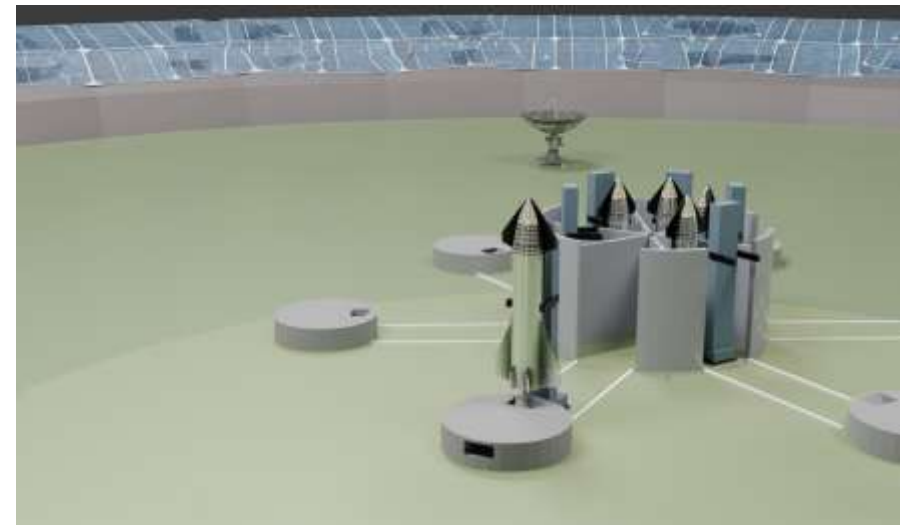
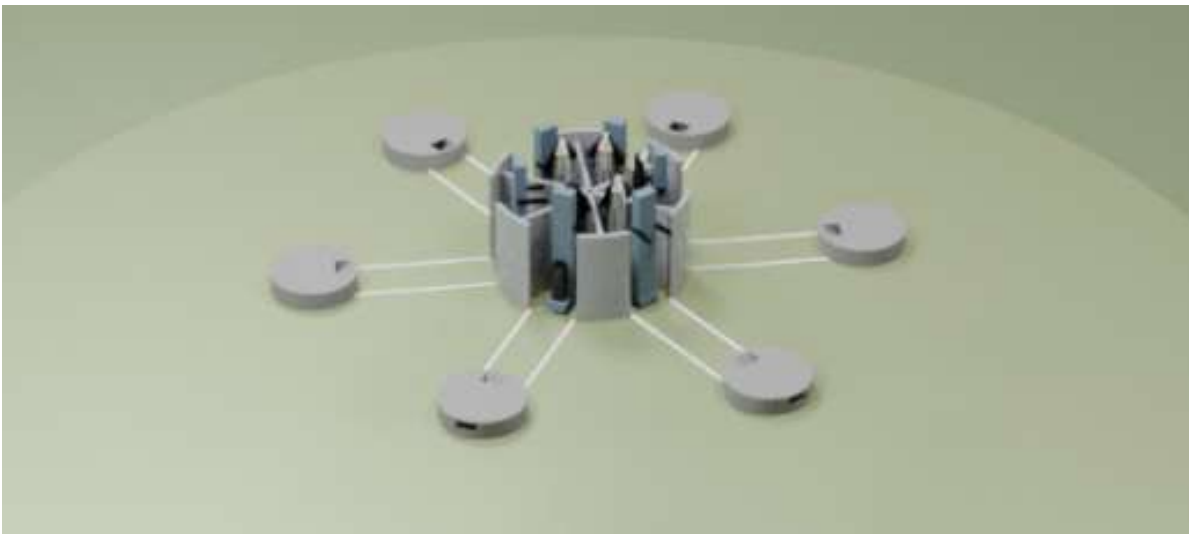


5.3.1 Port Automation

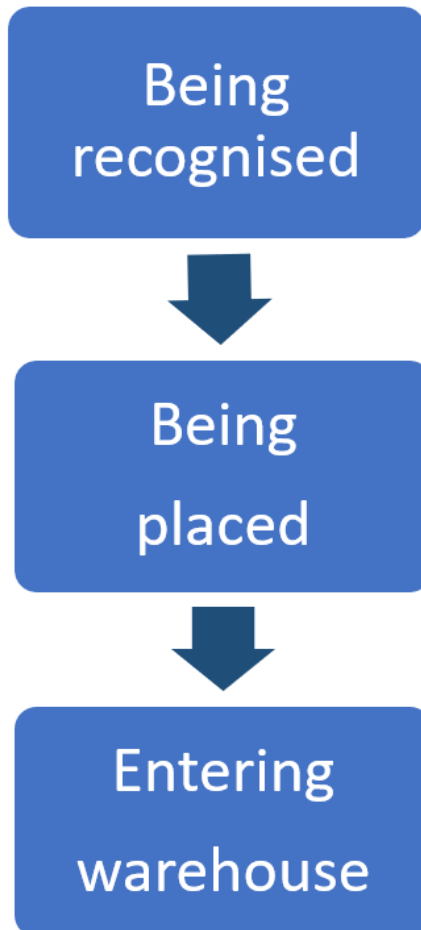
1 storage can store 6 rockets.

Rockets are semi-hidden in the storage, thus preventing sulfuric acid gas and violent storm wind.

Control tower base is used for carrying rocket.



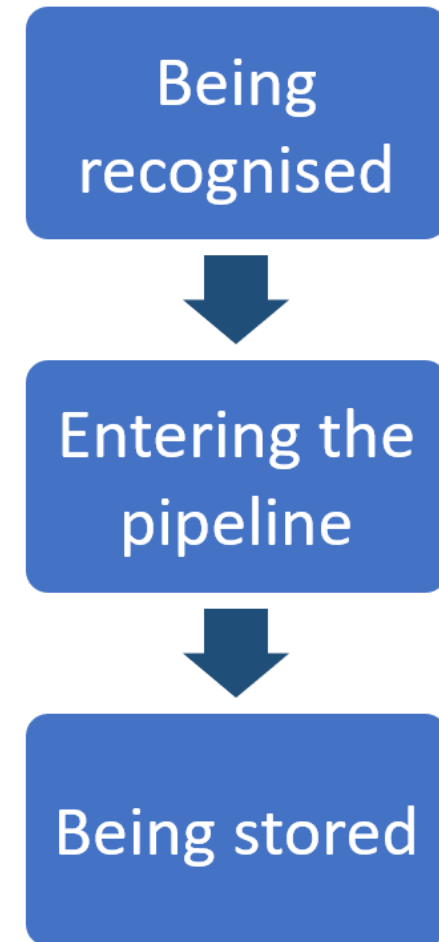
For dry ice:



For agricultural products:

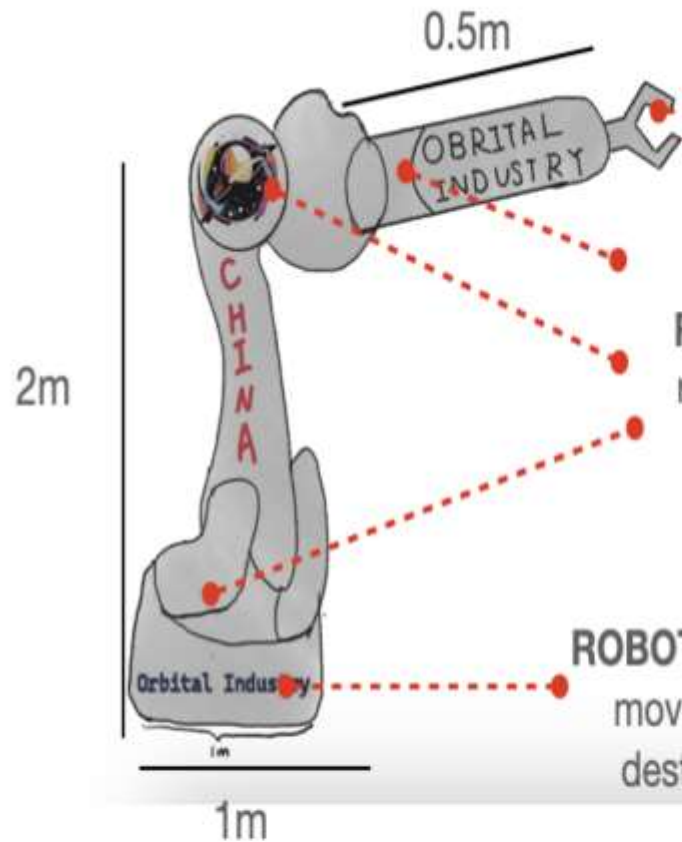


For liquid nitrogen:



5.3.3 Port Automation

PORT2-BOTTOM PORT



ROBOT CLAW-
pick up stuffs
from space ship

ROBOT JOINT:can
rotate 360 degrees
to take foods

ROBOT LEG:can
move to the
destination

Robots take cargos from spaceship



then to specific elevator



Then to the specfic places

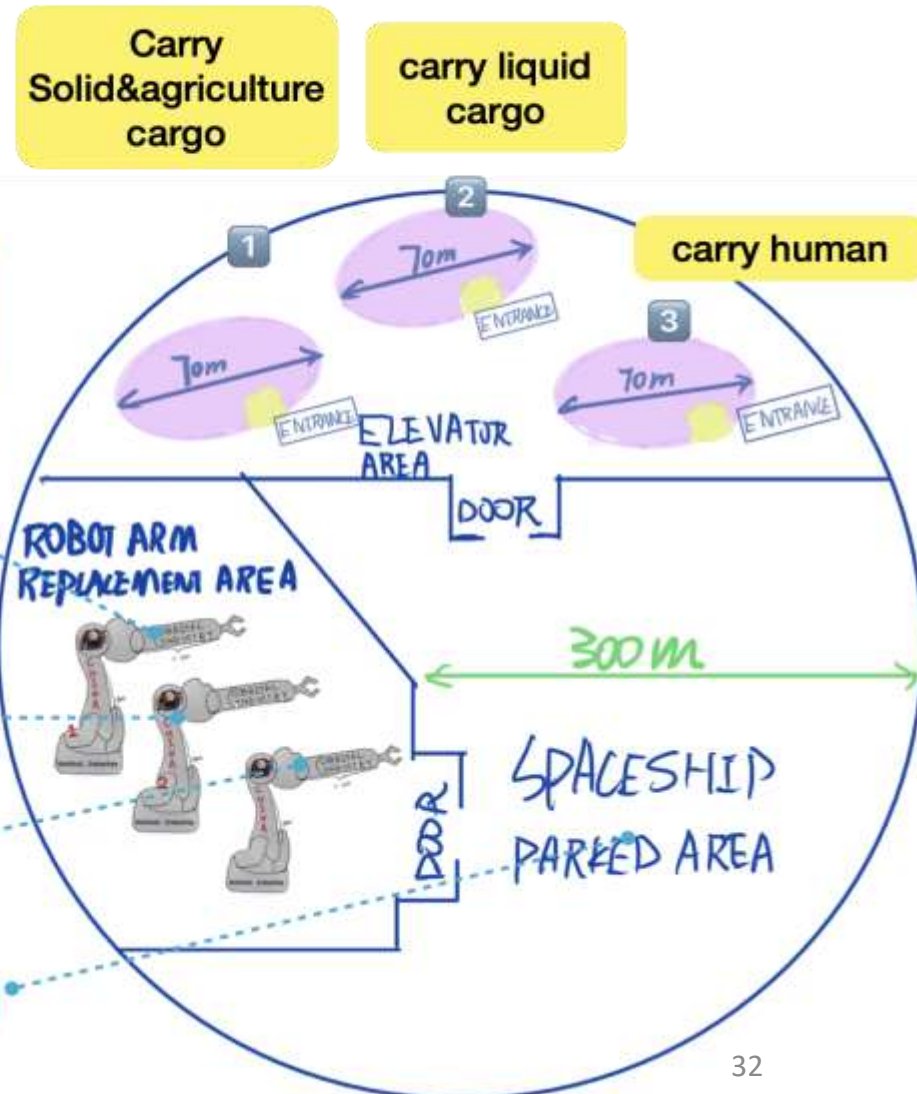
Port 2 data base
radius:300m

Robot arm 1-car
solid cargo

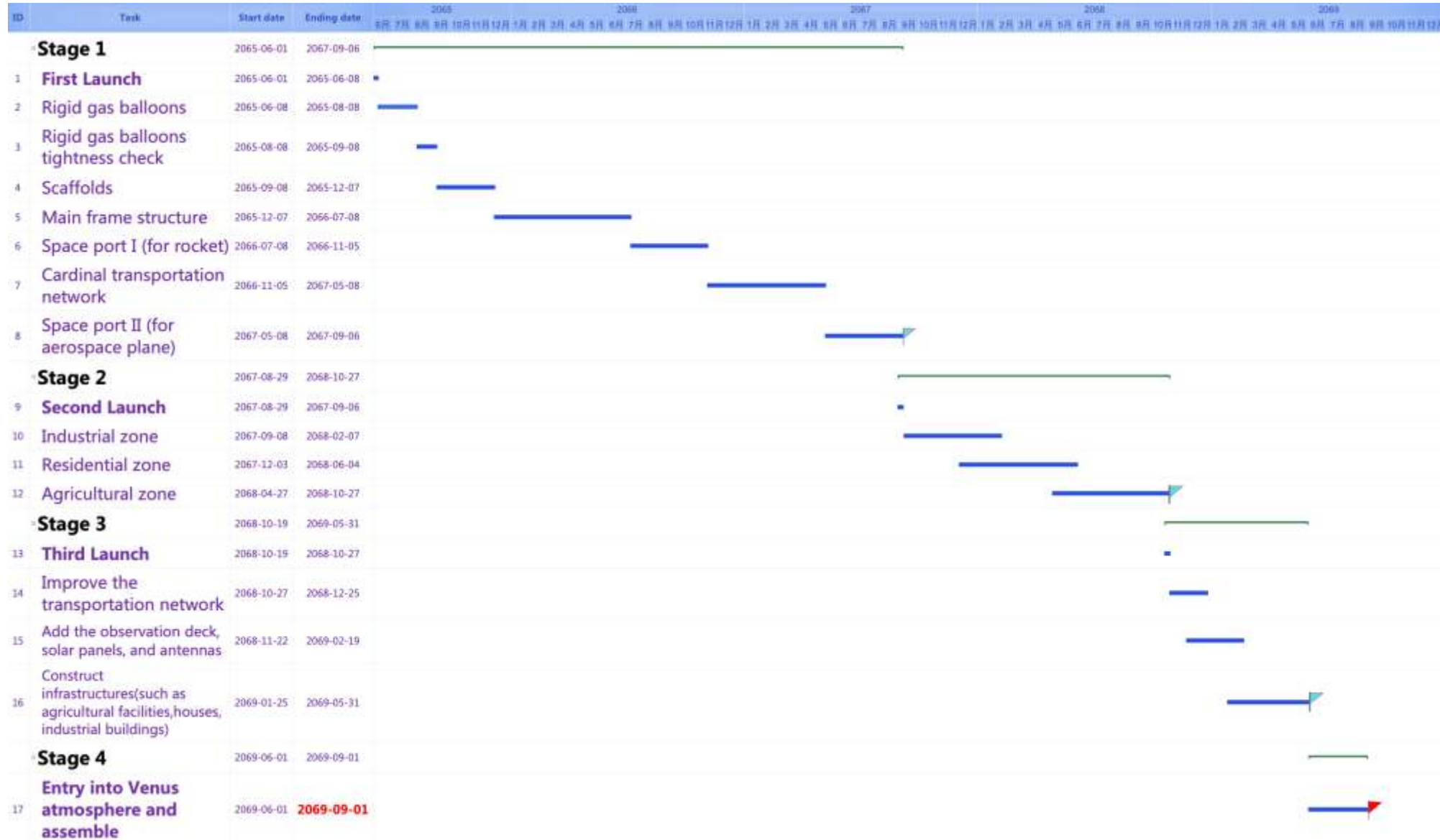
Robot 2arm -carry
agriculture cargo

Robot arm 3-carry
fluid cargo
(Bottled Liquid)

Can parked **approximatel**
spaceships



6.1 Schedule



6.2.1 Cost

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Process	First stage consturcture				Second stage Resident area		Third stage Infrastruct ure		Forth stage other facility			
Item	material	manufacturi ng and processing	transportation and assembly	manp ower +mac hine	construc- tion	expert	eleva tor	pipel ine	solar panel	agricul ture	airshi p	rocket
Quantity	0.95 billion kg	3.81*10^ 5m	0.95 billion kg	1.25 billion	90789m^ 2	1000	388 m	8491 44m	2558 256 m^2	392 m^2	25	10^7 RMB/ day
Unit Cost (RMB)	5000	10000	1500	1000	900	250000	1287	11.5 9	1248	3068. 12	7,600 ,000	21,000 ,000
Total Cost (billion RMB)	4750	9500	1425	1,250	0.08171	0.25	0.000 5	0.00 9842	31.92 5633	0.001 23	0.190	2.1

Total material expenditure during the construction phase: 6960.477205billion RMB

6.2.2 Revenue

35

a year's total revenue: **531.742** billion RMB

50 year's total revenue: **26,587.1** billion RMB

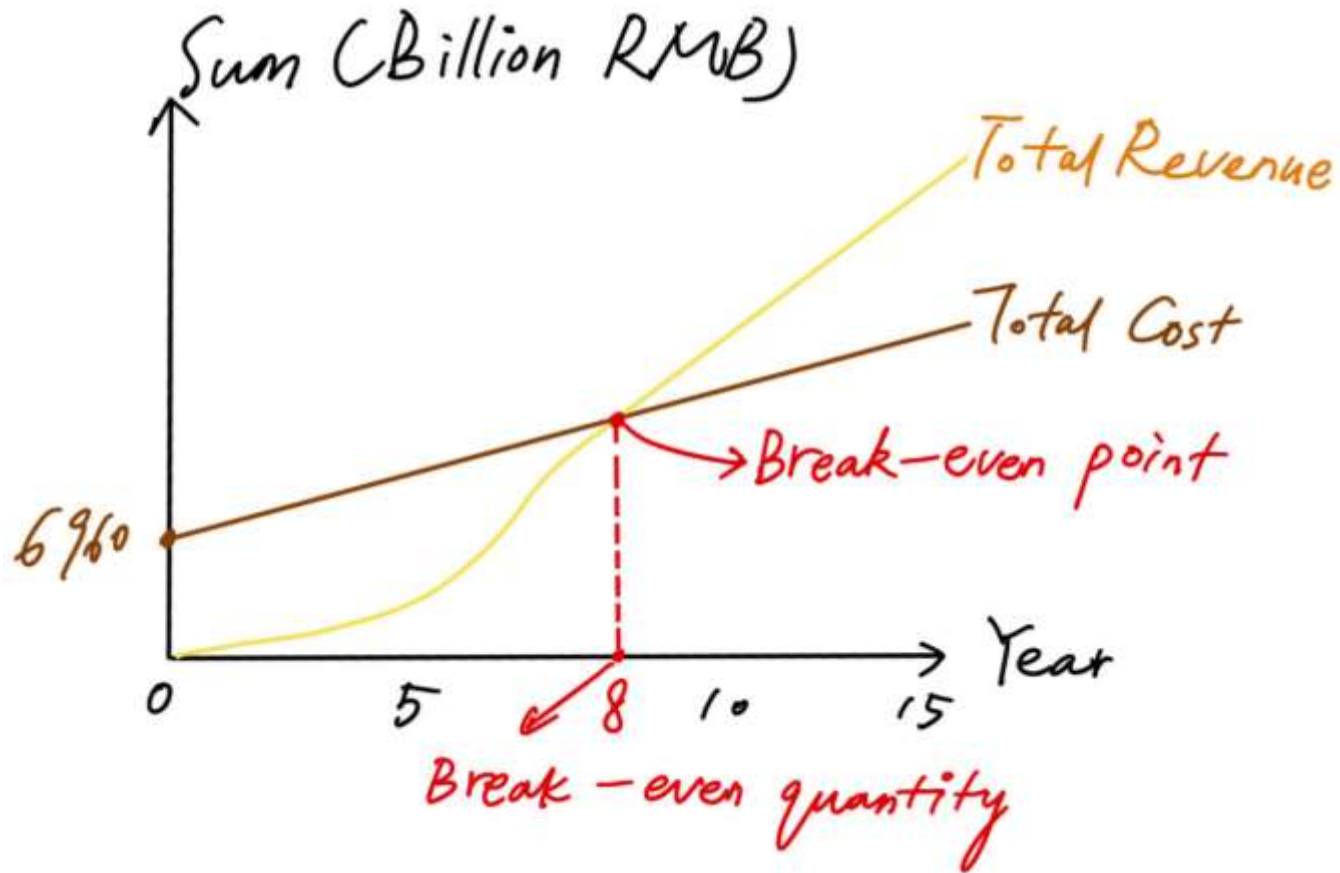
Category	Agriculture	People-inhabited					Transport
Item	crop income	school-tuition fee	cargo movement	travel	live	recreation	entrance fee
Quantity	392m ²	200	100,000	500	3000	182500	3500
Unit Cost (RMB)	100,000	5,000,000	100,000	10,000,000	50,000,000	2,000,000	100,000
Total Cost (billion RMB)	0.392	1	10	5	150	365	0.35

total cost: **10000** billion RMB

total net revenue: **5027.1** billion RMB

6.2.2 Revenue

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After comprehensive consideration of various factors, We calculated that the space city can achieve a balance between income and expenditure in the eighth year. This table shows the every year income and expenditure of Space City.



THANKS FOR
LISTENING